Predicting H5N1 Outbreaks Using Social, Search, and Environmental Data

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Abstract

Social media data has become a valuable new source of insight into the spread of diseases and public discussion of outbreaks and the response of their governments. Using data from Twitter/X, Meta Content Library, and Global Database of Events, Language, and Tone (GDELT), World Health Organization (WHO), and the Centers for Disease Control and Prevention (CDC), this study analyzes awareness, sentiment, and geographic trends related to the H5N1 avian influenza virus, commonly known as "bird flu". We apply a suite of natural language processing methods involving stigma classification, sentiment polarity, topic modelling, and time series analysis.

We demonstrate the effectiveness of NLP tools for digital epidemiologic analysis of H5N1 by identifying early warning signals in social media data prior to the announcement of an outbreak. While the model generates false positives and is not suitable as a standalone detector, it offers potential value as an auxiliary early-signal system for public health monitoring. Feature importance analysis highlights a rolling average of urgency to be a key predictor, consistent with prior research showing disease outbreaks correspond with a gradual buildup in early warning signals rather than sudden spikes in activity. Topic and sentiment analysis identifies major themes of discussion such as human cases, agriculture, wild bird populations, and government response, demonstrating the utility of NLP techniques for organizing and interpreting large scale social media discourse relating to public health.

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Chapter 1: Problem and Setting

1.1 Introduction

The H5N1 avian influenza virus, commonly known as "bird flu", is a highly pathogenic subtype of influenza A that primarily infects birds. The current outbreak has affected wild birds, poultry, and cattle across multiple continents. Despite the relatively low risk of human infection, the virus has garnered global attention due to high mortality in confirmed human cases and rising concern over its zoonotic transmission potential.

The recent spread of the H5N1 virus has intensified digital discourse on social media and online platforms, which have become primary sources of information for the public. This digital discourse is highly vulnerable to misinterpretation or manipulation, directly shaping public awareness and risk perception. These cognitive responses are crucial in determining individual and collective health behaviors during infectious disease outbreaks.

This paper explores the relationship between digital discourse and public responses to the H5N1 outbreak, focusing on how online communication affects awareness, risk perception, and the spread of information. This research may be used to inform public health communication strategies around infectious disease outbreaks and enhance understanding of public sentiment towards health topics in online environments.

1.2 Problem Statement

Existing research on the spread of H5N1 is substantial and developing, but the social dimension is underexplored. Current approaches to outbreak communication and surveillance often fail to integrate real-time digital discourse analysis, which is crucial for understanding how people perceive risk and respond to health advisories in the digital age. Without comprehensive strategies that incorporate digital communication dynamics, public health efforts risk being compromised by misinformation and misperceptions, potentially leading to inadequate outbreak awareness and risk mitigation behaviors. This gap threatens to exacerbate the spread of H5N1 and delay timely responses to emerging threats, ultimately increasing the risk of a more severe public health crisis. These limitations highlight the urgent need for surveillance models that integrate digital trace data alongside traditional epidemiological indicators.

1.3 Subproblems

- What is the temporal relationship between digital H5N1 discourse and official outbreak reporting? Do digital signals emerge before official announcements, concurrently with media coverage, or after official reporting becomes available?
- What communities/demographics are more likely to believe and share mis/information on H5N1?
- What are the sources of information on H5N1 (like public health agencies, CDC, special interest groups, independent actors)? How do these different sources emerge in the information landscape over time during an outbreak?
- On which platforms does H5N1 mis/information most commonly emerge and spread, and how do the patterns differ across platform types (social media, messaging apps, specialized forums)?
- What is the geographic correlation between digital signals and prevalence? For instance, are countries with higher H5N1 prevalence among livestock more likely to have more digital activity around H5N1?

1.4 Definition of Key Terms

- **H5N1**- Commonly known as avian influenza or "bird flu", a subtype of Influenza A that is highly contagious within and between bird populations. Has also been confirmed in other animals (such as cows) and in humans.
- **Prevalence**- The percentage or proportion of a population with a characteristic of interest, such as a disease.
- **Zoonotic**- Able to be transmitted from animals to humans
- **Digital discourse/signals**- Communication occurring primarily in digital spaces such as social media, messaging apps, text or video blogs, and websites. Can include text, images, or symbols (such as emoji).
- **Misinformation**- Information that is incorrect but is not shared with the intent to deceive, mislead, or harm- for example, unintentional errors in reporting, accidentally sharing a rumor or statistic that is untrue.
- **Disinformation**- Information that is incorrect and is deliberately shared with the intent to deceive, mislead, or harm- for example, hoaxes, deepfakes, conspiracies, or scams.
- **Natural language processing (NLP)**- A field of artificial intelligence that focuses on training computers to understand, interpret, and produce human-sounding language.

- **Sentiment analysis**- An NLP process used to interpret opinion or emotion in text based on identifying words commonly associated with various sentiments. Frequently used to classify sentiments as positive, negative, or neutral.

1.5 Significance of the Study

This study contributes to the growing field of digital epidemiology by evaluating how public discourse on platforms such as X (formerly known as Twitter), Facebook, and the Global Database of Events, Language, and Tone (GDELT) can serve as an early indicator of emerging public health threats like H5N1. By examining sentiment and post topic trends prior to official outbreak announcements, the study explores the predictive utility of digital signals in public health surveillance.

A key strength of this project is its analysis of regional and platform-level differences in how H5N1 is framed. These differences provide insight into the sociocultural shaping of disease narratives, which is essential for understanding how risk perceptions and media environments influence public health outcomes.

The project also identifies emotional and thematic patterns in H5N1-related discourse, particularly in comparison with Mpox and COVID-19. This comparative approach enriches the literature on health communication, stigma, and fear amplification. It highlights the importance of designing equitable and inclusive public messaging in the face of global health emergencies.

Ultimately, the findings from this research can inform targeted communication strategies that account for varying risk perceptions and media environments. This supports efforts by organizations such as the World Health Organization (WHO) and Centers for Disease Control and Prevention (CDC) to reduce stigma and enhance the effectiveness of health campaigns during emerging infectious disease outbreaks. Consequently, the study highlights the critical role that digital indicators can play in strengthening future outbreak preparedness frameworks.

Chapter 2: Theoretical and Conceptual Framework

2.1 Theoretical Background

This study is grounded in three interrelated theoretical frameworks that inform our understanding of how digital content reflects and shapes public discourse during infectious disease outbreaks: digital epidemiology, information disorder theory, and the Social Amplification of Risk Framework (SARF).

2.1.1 Digital Epidemiology

Digital epidemiology emphasizes the use of digital data sources- such as search queries and social media posts-to monitor and predict disease trends. Rather than relying solely on traditional surveillance systems, this approach leverages large-scale, real-time behavioral data to detect early indicators of public health concerns. Prior work has shown that search engine data can closely track disease prevalence, as demonstrated by Google Flu Trends (Ginsberg et al., 2009), and that social media platforms such as Twitter can serve as complementary sources for syndromic surveillance (Paul & Dredze, 2011; Salathé et al., 2012). This framework provides a foundation for interpreting digital activity surrounding H5N1 and for assessing whether online discourse precedes traditional outbreak reporting.

2.1.2 Information Disorder Theory

Information disorder theory provides a taxonomy for understanding the spread of false or misleading content during health crises. It distinguishes between misinformation (false but not intentionally harmful), disinformation (false and intended to cause harm), and malinformation (true but used maliciously), allowing researchers to classify digital content based on intent and veracity (Wardle & Derakhshan, 2017). This framework is particularly relevant in the context of zoonotic threats like H5N1, where rapidly evolving narratives on platforms such as Twitter and Reddit can shape public understanding and behavior. In this study, the framework supports the identification and categorization of misleading narratives within H5N1-related discourse across platforms.

2.1.3 Social Amplification of Risk Framework (SARF)

The social amplification of risk framework (SARF) posits that risks are not simply communicated but are actively interpreted and modified by social and media systems. These amplification or attenuation effects influence how the public perceives and reacts to hazards (Kasperson et al., 1988). In the case of emerging infectious diseases, emotional content- such as fear, anger, or humor- may heighten perceived risk, sometimes beyond the actual epidemiological threat. For this study, SARF provides a lens for interpreting how emotional expressions and thematic patterns within H5N1 discourse contribute to the amplification of perceived risk online.

These theoretical frameworks offer a robust foundation for examining how digital discourse not only reflects public sentiment and awareness but also shapes the social construction of health risk and misinformation during the early stages of infectious disease emergence. Moreover, they guide the analytical approach used in this study and support the interpretation of temporal, emotional, and geographic patterns within digital H5N1 discourse.

2.2 Literature Review

2.2.1 H5N1 Overview

H5N1 is a subtype of highly pathogenic avian influenza that primarily infects birds but has also crossed species barriers, causing outbreaks in wildlife, farm animals, and occasional human infections. While no human-to-human transmission has been observed, H5N1 remains a serious pandemic threat due to its high mortality rate and its potential to acquire adaptations quickly through reassortment with mammalian viruses. These concerns, combined with widespread outbreaks in poultry and cattle since 2021, highlight the importance of monitoring H5N1 as an emerging disease and developing rapid detection and response strategies (Krammer et al., 2025).

2.2.2 Use of digital communications in disease surveillance and early detection of outbreaks

Several studies have demonstrated the utility of digital signals for real-time disease surveillance. Ginsberg et al. (2009) introduced Google Flu Trends, using aggregated search queries to estimate influenza activity. More recently, studies have applied Twitter data (Paul & Dredze, 2011) and Reddit discussions (Sarker et al., 2020) to detect outbreaks such as COVID-19 and Mpox. These works provide foundational evidence that online platforms can serve as alternative surveillance systems, albeit with challenges related to representativeness and noise.

Multiple studies found an association between an increase in avian flu-related Google search trends and X (formerly Twitter) posts and official outbreak declarations (Soltani et al., 2025; Robertson and Yee, 2016), with Soltani even finding that increases in online activity tend to predate official reports of outbreaks by one to two weeks. While Soltani concludes that this association indicates online discourse could be used for early outbreak warnings, Robertson is more cautious about drawing this conclusion, since social media posts and searches for information could be reactive to early reports of individual cases in a region, rather than predicting an outbreak.

Paul and Dredze (2011) developed the Ailment Topic Aspect Model (ATAM) to extract structured health information from over 1.6 million tweets, distinguishing between ailments, symptoms, and treatments. Their model showed a Pearson correlation of 0.96 between the proportion of tweets about influenza and official CDC data, illustrating the strong potential of social media as a complementary tool for real-time syndromic surveillance.

In current research, there is not strong evidence that online discourse alone can be used to indicate signs of an outbreak- it is possible that increased activity is more a sign of increased local reporting leading people to search out information about the virus. There is a lack of comparison between early case reports or incidence rate and searches or social media posts (ex. Do social media posts start occurring before cases are officially reported or just before an outbreak- indicating many cases- is declared?), which would be necessary to determine whether digital discourse is a meaningful predictor or just a proxy for early case reporting.

2.2.3 Common methods for sentiment analysis and topic modelling

The most commonly used techniques were Latent Dirichlet Allocation (LDA) for topic modelling, to identify distinct themes that frequently occurred in discourse around COVID-19 and H5N1, and sentiment analysis to identify emotions of posts as a measure of polarization (Cooper et al, 2025; Gupta et al., 2022; Jafarinejad et al., 2021; Jang et al., 2020).

Jafarinejad et al. (2021) and Gupta et al. (2022) both explore how discourse evolved during the COVID-19 pandemic- Jafarinejad through dynamic topic modelling (following shifts in topics over time rather than viewing them statically, as LDA does) and concept drift detection (pinpointing when shifts in sentiment occur to compare to recent public health events). Gupta used a mathematical model to simulate social networks using YouTube comments and compared how polarized these networks were at various time points, finding that polarization increased as the pandemic continued. While Gupta’s simulation of “social networks” using disconnected YouTube comments (since they did not include replies, there is no way to know if the commenters ever had any interaction with each other) seems to be of little practical use, it offers a perspective on how YouTube comments started from more polarized positions later in the pandemic, which would theoretically make it harder for individuals with differing opinions to come to a consensus.

Most studies use traditional machine learning (ML) methods like sentiment analysis and

LDA. There is an opportunity to use more advanced large language models (LLMs) like BERT or GPTs to synthesize data at a more nuanced level than ML models which largely rely on identification of keywords. Soltani et al. (2025) use DistilBERT to filter out less relevant X posts and, Medford et al. (2020) uses tools such as VADER, BERT-based sentiment classifiers, and emotion lexicons to assess sentiment shifts over time, but none of the other studies identified thus far use contextual language models to analyze the data.

Current studies mostly focus on text, without looking at other potentially useful signals like emoji, images (ex. memes), or videos which are frequently used to communicate information (or misinformation) about public health events.

2.2.4 Information, misinformation, and disinformation detection and patterns

Mis/disinformation are common during public health crises, but can be hard to detect and verify. However, topic modelling may provide a useful tool for flagging mis/disinformation in social media posts by identifying commonly used terms ("hoax," "plandemic," "scam," "election") that indicate conspiracy and mistrust (Cooper et al., 2025).

During COVID-19, the spread of health misinformation online was identified as a critical threat to public health. Research by Kouzy et al. (2020) found that nearly one-third of COVID-related tweets contained misinformation. Similarly, Islam et al. (2020) mapped disinformation networks and their impact on vaccine hesitancy. These studies underscore the importance of early detection and classification of false content, especially in the initial stages of an outbreak when uncertainty is high.

Cinelli et al. (2020) analyzed the platform-specific dynamics of COVID-19 misinformation and modeled its spread using epidemiological concepts like the reproduction number (R_0). Their findings revealed that low-credibility information consistently exhibited R_0 values above 1, with users of the social media platform Gab engaging with such content nearly three times more than with credible sources, highlighting the structural amplification of misinformation across platforms.

Most studies focused on sentiment and polarization rather than trying to identify the role of dis/misinformation. This is a limitation in current research, since concordance of opinion may not be the desired outcome. In Gupta et al. (2022), for instance, their model tracks how many iterations each simulated network takes to reach a “middle ground” (sentiment scores near the median instead of at the extremes). However, if sentiment is influenced by true/false information, coming to an agreement with potentially faulty information may be more harmful to public health than polarization is.

2.2.5 Sociodemographic and Regional Factors

Studies by Bridgman et al. (2020) and Cinelli et al. (2021) found that misinformation exposure and trust in institutions vary significantly across regions, age groups, and education levels. Regional inequalities in digital access and cultural trust norms shape how health information is received and interpreted. This suggests that the same digital signals may mean different things in different communities.

Twitter/X and Google search trends are the most commonly used platforms in digital surveillance research due to their widespread use and ease of obtaining data (Cooper et al, 2025; Jafarinejad et al., 2021; Jang et al., 2020; Soltani et al., 2025; Robertson and Yee, 2016), but they may not be representative of the general population. Only Jafarinejad et al., (2021) (Twitter, Instagram, Telegram) and Gupta et al. (2022) (YouTube) used other social media sites that may capture a different portion of the population.

Most of the studies found so far focus on North America (particularly the US and Canada) and on English-language data due to availability and language limitations of the researchers (Cooper et al, 2025; Gupta et al. ,2022; Jang et al., 2020). Jafarinejad et al. (2021) uses exclusively Farsi-language posts from Iran, and Soltani et al. (2025) uses tweets and search trends from multiple countries and in multiple languages (English, Spanish, French, Japanese, Italian), making them more geographically and linguistically diverse than other studies. Soltani uses NLLB-200, a translation model developed by Meta, to translate posts into English for analysis. While this increases the amount of data they can collect, there is no assessment of how accurate these translations are. In addition, there is a lack of cross-region comparative studies which could provide useful information about how different social, cultural, and demographic contexts engage in online discourse and respond to mis/information about outbreaks.

2.3 Contribution and Distinctiveness of This Study

In the context of the ongoing global H5N1 outbreak as a public health threat, this study aims to draw attention to the intersection of digital epidemiology, digital social media and traditional analytics, and public health. While previous studies on H5N1 have predominantly focused on virological properties, zoonotic transmission pathways, and epidemiological modeling, relatively less has been explored regarding the digital content from the general public and media outlets, i.e., how public sentiment and opinion form online in real time during zoonotic spillover events.

2.3.1 Key Differentiators from Prior Research

Unlike previous work, which tends to discuss social media and online data as retrospective predictors, this study proactively explores their role as forward predictors. Furthermore, this study attempts to investigate whether online discussion, via social media updates, search requests, and news articles, predicts formal reports of outbreaks and under what emotional and thematic patterns change coincides with epidemiologic events in synchronization. This forward-looking aspect brings sentiment analysis and geospatial-temporal correlation into a coherent framework and provides a more comprehensive understanding of outbreak dynamics.

In addition to temporal and thematic differentiators, sociodemographic and cultural filtering mechanisms play a critical role in shaping how digital discourse around infectious disease outbreaks is formed and interpreted. Bridgman et al. (2020) found that individuals who relied more heavily on social media—particularly those with politically conservative orientations—were significantly more likely to endorse COVID-19 misperceptions. This relationship was further moderated by demographic factors such as age and education level. These findings underscore that digital expressions of fear and skepticism are often entangled with deeper ideological and structural dynamics. Furthermore, as posited by the Social Amplification of Risk Framework (Kasperson et al., 1988), risk signals such as H5N1 outbreaks are not simply transmitted but are mediated and amplified through local cultural norms, institutional trust, and media ecologies. Taken together, these considerations highlight the importance of regionally sensitive modeling when analyzing emotional volatility and narrative shifts in the digital public health landscape.

2.3.2 Exclusive Contribution of This Research

Our team supplements the existing literature with an interdisciplinary framework that integrates Natural Language Processing (NLP), Agentic AI with Large Language Models (LLMs), and regional sensitivity modeling to analyze online participatory reactions to H5N1. Our work sets up five specific, falsifiable hypotheses across emotional volatility, and regional sensitivity axes to enhance public health surveillance systems. Using human-in-the-loop and AI-based classifiers, we provide a rich and scalable solution for outbreak signal detection and real-time narrative tracking.

While prior studies such as Paul and Dredze (2011) employed probabilistic models like the Ailment Topic Aspect Model (ATAM) to extract structured health signals from social media data, they primarily focused on classifying tweets by ailments, symptoms, and treatments. However, these models generally lacked temporal alignment with real-world epidemiological events. In contrast, our study integrates emotional and topical signals along a synchronized outbreak timeline. This dual-axis design—tracking what people feel and discuss, and when—enhances both interpretability and forecasting capacity, offering a novel framework for digital discourse analysis in epidemic contexts.

2.3.3 Why This Study is Relevant and Necessary

This study is necessary and timely since public health systems are constantly challenged by the ever-evolving spread of digital discourse, especially during the early stages of an outbreak. Given that public behavior is strongly influenced by digital media, understanding panic propagation can be as critical as tracking the spread of the virus itself. Our research bridges an important gap by providing tools and analytical frameworks to trace, interpret, and anticipate emerging patterns of digital stigma. This allows us to gain early insight into how communities discuss and respond to unfolding public health events. The core innovation of this study lies in its approach to treating digital signals not merely as reflections of public sentiment but as actionable intelligence for future pandemic preparedness.

Beyond academic contributions, this study also responds to a growing need for tools that can monitor the consistency and effectiveness of public health communication in real time. For instance, Bulut and Poth (2022) conducted a sentiment analysis of 131 official COVID-19 briefings in Alberta, Canada, and found that despite dramatic shifts in case numbers and policy stringency, the emotional tone of the briefings remained remarkably stable. Their findings suggest that sentiment scores can serve as reliable indicators of message coherence and institutional trust during health crises. By analyzing the volatility and direction of digital discourse, our study contributes to the timely detection of shifting public narratives and supports more adaptive and targeted policy responses.

Chapter 3: Methodology

3.1 Research Objectives and Questions

The project seeks to identify whether online streams of data in the form of social media updates and published news articles can serve as early warning indicators of public awareness and information spread during zoonotic threats such as H5N1. With the analysis of temporal, emotional, and thematic characteristics of online discourses, the project intends to complement conventional epidemiological surveillance systems with timely public insights.

The project further aims to:

- Explore how discourse signals emerge in the early phases of online discussion on H5N1.
- Quantify whether particular emotional trends (e.g., fear, uncertainty, trauma response or humor) dominate conversation before official public health announcements.
- Determine whether digital content trends differ by geographical regions and sites.

The following research questions will guide our intended study:

- Can digital signals act as early indicators of awareness?
- Do noticeable increases in H5N1-related digital content occur prior to official reporting of public health concerns or outbreaks?
- How do emotional cues such as fear, urgency, denial, or humor evolve as the public health narrative unfolds?

These questions are in alignment with key subproblems such as outbreak response latency, digital epidemiology, and public trust in health communications.

3.2 Research Hypotheses

Building on previous research in digital surveillance and sentiment analysis related to Mpox and COVID-19, we propose the following testable hypotheses. Each hypothesis is designed to be empirically measurable using available datasets such as timestamped social media content, search engine query logs, and news coverage records.

Hypothesis 1: Awareness Signal Hypothesis
Spikes in the volume of H5N1-related search queries and social media mentions will occur

at least three days prior to the first official outbreak announcement by the World Health Organization (WHO) or Centers for Disease Control and Prevention (CDC).

- **Variables**: Timestamped search queries and social media posts; date of official outbreak announcement
- **Test**: Time lag analysis between digital activity and official reports
- **Data**: Social media logs, Google Trends data, WHO/CDC announcements
- **Metric**: Number of days between digital signal peak and official report
- **Falsifiability**: If signal spikes are not statistically significant before official reports, or appear afterward, the hypothesis can be rejected.

Hypothesis 2: Emotional Spike Hypothesis
Social media posts expressing uncertainty or fear will peak in the pre-confirmation phase and decline following official public health communications.

- **Variables**: Timestamped social media posts; emotion labels classified via LLM or emotion classifiers
- **Test**: Time-series analysis of emotion label frequency
- **Data**: Text-based emotion classification outputs from LLMs or pre-trained models
- **Metric**: Temporal trend in emotional content (e.g., fear, anxiety, uncertainty)
- **Falsifiability**: If emotional signals do not peak prior to confirmation, or peak after, the hypothesis can be rejected.

Hypothesis 3: Regional Sensitivity Hypothesis
Spikes in fear-related or speculative digital content will be more immediate and pronounced in regions with limited prior exposure to zoonotic disease outbreaks, compared to regions previously impacted by Mpox or COVID-19.

- **Variables**: Region or country; emotion labels; timestamp; prior exposure index
- **Test**: Temporal and geospatial comparison between regional clusters
- **Data**: Region-tagged digital content; historical exposure metrics
- **Metric**: Magnitude and rate of emotional or speculative content by region
- **Falsifiability**: If patterns are similar across all regions, or if high-exposure regions show stronger reactions, the hypothesis can be rejected.

When considered in combination, these hypotheses contribute to the theoretical and practical advancement of digital public health surveillance, with implications for early detection, public health communication strategies, and region-specific response strategies in the context of zoonotic threats.

3.3 Conceptual Framework

This study presents a conceptual framework for analyzing whether digital signals can function as early indicators of public awareness during zoonotic outbreaks such as H5N1. The framework connects theoretical foundations with empirical procedures to investigate how social media posts, search engine trends, and online news content might precede official public health announcements.

The framework consists of four main components: digital data inputs, theoretical perspectives, analytic processes, and interpretive outputs. It begins with the collection of relevant digital content from platforms such as Twitter, Facebook, and GDELT. These inputs are then analyzed through structured methods focusing on temporal, emotional, geographic, and thematic dimensions. Based on the results, potential early warning indicators are derived and compared with official communications from health agencies. This process includes a feedback mechanism that allows ongoing refinement as new discourse and reports emerge.

The framework builds upon three theoretical perspectives that offer insight into the function of digital content during public health events.

Digital epidemiology supports the use of online behavioral data, such as search engine queries and social media activity, to detect early signs of public awareness related to health threats. This approach enhances traditional disease surveillance by incorporating large-scale and real-time public reactions.

The social amplification of the risk framework explains how risk perception is shaped by communication patterns and media environments. Emotional content in online discourse, such as fear, anger, or humor, can intensify or reduce public concern, sometimes beyond the level warranted by epidemiological data.

These three perspectives form the theoretical basis for the study's analytical components. They guide the measurement of timing in digital signals, the detection of emotional patterns, and the identification of regional differences in how digital discourse unfolds. This foundation supports the study's core hypotheses concerning awareness signals, emotional response, and regional sensitivity during the H5N1 outbreak.

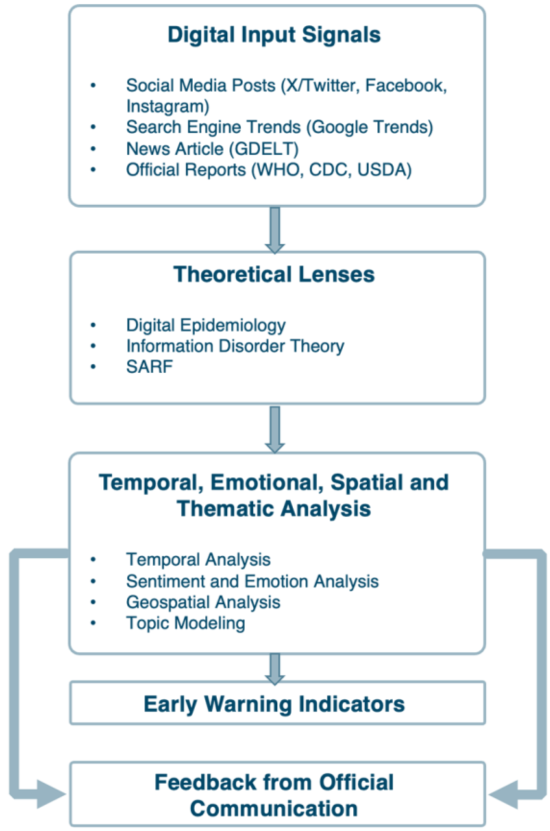


Figure 1: Conceptual Framework for Digital Signal Analysis During H5N1 Outbreaks

The framework shows how digital signals from sources such as social media, search trends, and online news are processed through theoretical and analytic layers. These layers produce early warning indicators related to awareness and emotion. These indicators are then compared with official communications through a feedback loop that helps refine the analysis over time.

As illustrated above, the process begins with the collection of time-stamped and geo-tagged digital content. These data are analyzed using methods aligned with theoretical perspectives to identify meaningful signals. The resulting indicators are evaluated in the context of official outbreak announcements, allowing the model to adjust its interpretations based on verified events and institutional responses.

This conceptual framework offers a structured approach for examining whether public online discourse can serve not only as a reflection of awareness and emotion, but also as a proactive tool for early detection of health threats. By aligning theory with measurable indicators, it contributes to the growing field of digital epidemiology and supports the development of timely, evidence-informed public health strategies.

3.4 Research Design

This study adopts a mixed methods modeling framework to examine whether digital discourse surrounding the H5N1 outbreak can serve as a predictive and spatial signal prior to official public health reports. The design combines natural language processing (NLP), time-series analysis, and geospatial modeling to link online activity with epidemiological developments.

3.4.1 Methodological Overview

Our design integrates three core methodological layers:

- **Natural Language Processing (NLP):** To identify, classify, and quantify themes and emotional tones in H5N1-related discourse.
- **Time Series Forecasting:** To detect signal peaks in digital discourse that may precede official outbreak reports.
- **Spatiotemporal Analysis:** To assess how discourse intensity aligns with geographical regions of reported outbreaks.

This approach allows for a dynamic comparison of public reaction across time and space in relation to official epidemiological events.

3.4.2 Analytical Pipeline

The following steps outline the sequence of procedures in our research design:

1. **Data Harmonization and Ingestion**
    - Use SOMAR-compliant ingestion pipelines to collect and align data from APIs (Twitter/X, Meta, GDELT).
    - Convert all inputs into time- and region-tagged units.
    - Normalize engagement metrics (likes, shares, search volume) for comparability.
2. **Signal Construction**
    - Build time series capturing H5N1-related tweet volume, news article frequency, and search trends.
    - Align these with official outbreak reports from WHO and CDC.
3. **Predictive Analysis**

- Evaluate whether digital stigma trajectories anticipate outbreak declarations by applying a time-aware logistic regression model incorporating lead–lag emotional features.

4. **Semantic and Emotional Modeling**
    - Apply BERTopic and DistilBERT to extract dominant themes and emotional tones.
    - Classify content into neutral, alarmist, and misinformation-likely categories using pretrained models and human-validated labels.
5. **Geospatial Analysis**
    - Aggregate discourse volume and tone by region.
    - Use Moran's I to assess spatial clustering and Pearson/Spearman correlations to compare digital discourse with outbreak prevalence.

3.4.3 Evaluation Metrics

To evaluate and assess how well our models perform, including those used for sentiment analysis and topic modeling, we used a set of evaluation metrics that are specifically chosen to fit the goals and structure of each model. This helped us measure both the accuracy and reliability of our models in real-world scenarios.

Below are the different categories of models used in this study. For each category, we outline the evaluation metrics and success criteria.

Classification Models (Sentiment, Emotion Detection)
We utilized the following evaluation metrices for binary and multi-class classification tasks such as sentiment classes, emotion detection etc.:

- **Accuracy:** Proportionality of correctly classified instances across all classes.
- **Precision:** Ability of the classifier not to label a negative sample as positive.
- **Recall (Sensitivity):** Ability to find all the relevant cases within a dataset.
- **F1 Score:** Harmonic mean of precision and recall; particularly useful for imbalanced classes.
- **ROC-AUC:** For binary classification tasks, we will include the Area Under the Receiver Operating Characteristic Curve to assess the model's discrimination capability.
- **Confusion Matrix:** To visually inspect class-wise misclassification patterns, particularly useful in imbalanced datasets.

Success Criteria: We define success as achieving an F1 score greater than 0.75 for high-stakes classes based on a manually annotated validation set of smaller set.

Time-Series and Early Indicator Detection (Search/Social Volume vs. Outbreak Data)
For aligning digital signal trends with epidemiological events:

- **Granger Causality Test:** To assess predictive relationships between online signals and reported cases.
- **Lag Correlation:** We used Lag Correlation methods to evaluate temporal alignment of signal peaks with case reports from WHO and CDC.

General Success Criteria
We considered a model successful based on the following criteria:

- Consistently outperforms naive baselines such as random or majority-class classifiers.
- Demonstrates high generalizability across platforms such as X/Twitter, Facebook, and GDELT.
- Captures early indicators of public sentiment shifts prior to epidemiological confirmation.

3.4.4 Integration with Hypotheses

The analytical procedures implemented in this study are designed to directly correspond to the hypotheses presented in Section 3.2. Each hypothesis is associated with specific data sources, modeling techniques, and evaluation criteria. The links between hypotheses and methods are outlined below.

Hypothesis 1: Awareness Signal Hypothesis
This hypothesis posits that significant increases in digital discourse, such as social media activity, search trends, and news coverage related to H5N1, occur prior to official outbreak announcements. To evaluate this, time series will be constructed using timestamped digital signals and aligned with announcement dates provided by public health agencies, including the WHO and CDC. Analytical methods include Granger causality testing and lagged cross-correlation, which will assess whether digital signals precede official reporting in a statistically meaningful way.

Hypothesis 2: Emotional Spike Hypothesis
The second hypothesis suggests that expressions of fear, uncertainty, and related emotions are more prevalent during the early stages of public discourse, particularly before official confirmation of outbreaks. Following confirmation, these emotions are

expected to decline. This hypothesis will be examined by applying emotion classification models to text data, allowing for the construction of emotion-labeled time series. Temporal trends will be analyzed to determine whether emotional intensity decreases after public health announcements, as hypothesized.

Hypothesis 3: Regional Sensitivity Hypothesis
This hypothesis proposes that regions with limited prior experience of zoonotic outbreaks will exhibit earlier and more intense fear-related or speculative discourse compared to regions that experienced Mpox or COVID-19. Content will be geotagged or inferred by region and categorized by emotional tone. Regional differences will be analyzed using spatial correlation techniques, including Moran's I and comparative trend analysis, to determine whether the timing and intensity of discourse vary as predicted.

The integration of time series analysis, semantic modeling, and spatial statistics within the overall research design allows for systematic testing of each hypothesis. By aligning modeling procedures with hypothesis structure, this study aims to generate empirical findings that contribute to the growing body of literature on digital epidemiology, public health communication, and early warning systems.

3.4.5 Technical Tools and Implementation Environment

We used a modular and scalable technical architecture that supports API provided data ingestion, processing, and machine learning analysis of online conversation during infectious disease outbreaks. Implementation of this project utilizes widely-used open-source tools, machine learning libraries, and collaboration platforms to facilitate reproducibility, flexibility, and team productivity. The table below categorizes the tools we used during the implementation phase of the project.

Tool	Description
Python	Programming Language
Github	Code Version Control and Collaboration
MLflow	Model/Experiment Management
Jira	Project Management
Huggingface	Language Models
Tensorflow/Keras/Pytorch	Deep Learning Framework
Data APIs (SOMAR, Reddit, Twitter)	Data Acquisition
Google Collab/Drive	Collaborative Cloud Compute w/GPU
Delta or Iceberg	Open Table Format
Apache Spark	Analytical engine
Apache Parquet	Open File Format

Table 1: Environment, tools, programming languages and frameworks used to support

3.5 Data Collection and Storage of Data

To support the analytical goals of this study, particularly the detection and characterization of public discourse related to H5N1, a diverse set of data sources was collected from both structured and unstructured platforms. The data included social media posts, search trends, news articles, and official epidemiological reports which contain temporal and geographic metadata necessary for time-series and spatial analysis.

3.5.1 Types of Data Used

- **Structured Data**: Includes epidemiological surveillance data such as outbreak case counts, situation reports, and alert timelines from public health agencies (e.g., WHO, CDC). Additional structured variables include engagement metrics from social media (e.g., likes, retweets, shares) and search indices.

- **Unstructured Data**: Consists of free-text content from social media posts, global news articles, and user-generated commentary. These data are used to extract sentiment, emotion, and thematic content.

- **Time Series Data**: All datasets are timestamped, allowing for longitudinal alignment of digital discourse trends with outbreak events.

- **Geospatial Data**: Many data sources include location metadata (e.g., geotagged tweets, regional search trends), which support regional sensitivity modeling.

3.5.2 Data Sources and APIs

The following sources and platforms are used to gather relevant data:

- **Twitter/X API**: Used to collect posts related to H5N1 through keyword filtering. Metadata such as post time, user location (when available), and engagement metrics are included.

- **Meta Content Library API (Facebook and Instagram)**: Enables access to public posts containing relevant keywords. Geographic diversity is emphasized to capture regional variance in discourse.

- **GDELT (Global Database of Events, Language, and Tone)**: Supplies global news coverage related to H5N1, tagged by location, sentiment, and event type.

- **WHO and CDC datasets**: Offer official outbreak timelines, case counts, and regional risk levels for reference and alignment.

3.5.3 Data Governance and Standardization

A SOMAR-based framework (Source, Observation, Metadata, Annotation, and Representation) will guide data ingestion and harmonization:

- **Source**: Provenance of data collection is documented (e.g., API, archive, scrape).
- **Observation**: Captures the original behavioral or textual signal (e.g., post, article, search).
- **Metadata**: Includes temporal, spatial, and engagement-related attributes.
- **Annotation**: Text data are labeled using machine learning and human validation to identify emotion and sentiment.
- **Representation**: Datasets are transformed into a standardized format for time-series modeling and geospatial analysis.
- **Data Provenance and Lineage:** Data lifecycle is retained and captured through data provenance and lineage implemented through data versioning. We will implement persisted data versioning capabilities with open-source table formats such as delta or iceberg in an attempt to retain data lifecycle.

3.5.4 Preprocessing Procedures

- **Language Filtering**: Only English-language content is retained for analysis to ensure compatibility with available NLP tools.
- **Normalization and Cleaning**: Datasets undergo de-duplication, noise filtering, and spam removal. Engagement metrics are standardized across platforms.
- **Topic Modeling Preparation**: Texts are embedded using TF-IDF and Non-negative Matrix Factorization (NMF).
- **Sentiment and Emotion Tagging**: VADER is used to classify posts into lexicon-based emotion categories.
- **Metadata Integration**: Engagement signals (e.g., likes, search volume) are used as weights in modeling, and all records are temporally aligned with outbreak timelines.

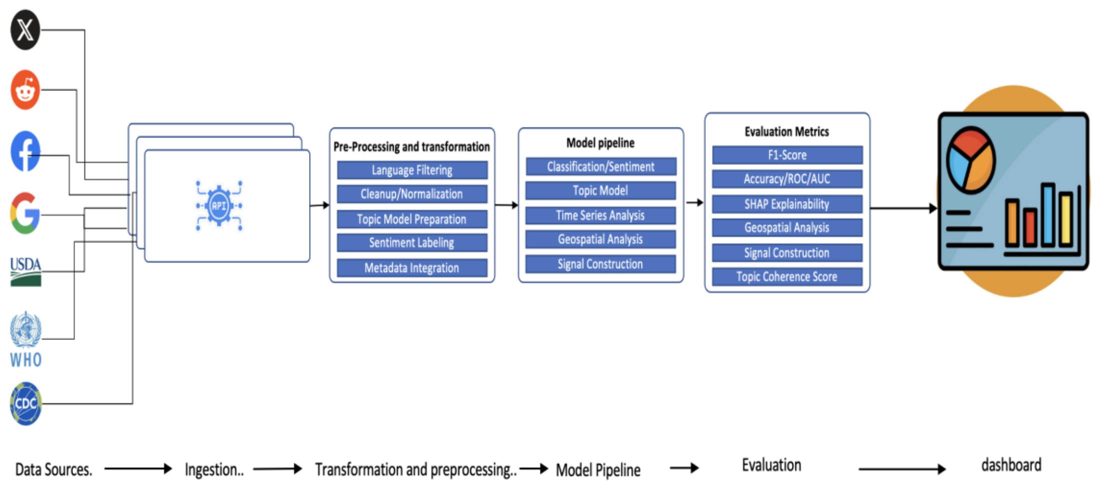


Figure 2: Data ingestion, pre-processing and model lifecycle pipeline

3.6 Sample and Sampling Procedures

The study uses a multi-source sampling approach in collecting and analyzing digital data and epidemiological data for the ongoing H5N1 outbreak. Sampling design is aimed at capturing temporal dynamics and thematic breadth within public debate to enable meaningful comparison with officially reported outbreak events.

Digital Discourse Data Sampling

We sampled digital content from several platforms known to contain feed of public health related text:

- **Twitter/X:** Using a combination of access to APIs and academic datasets, we extracted tweets containing "H5N1," "bird flu," "avian influenza," and their variations. We identified a sample frame range based on the data we are able to acquire through the API access process. This temporal sample frame range had an emphasis on periods of increased case reporting and media attention.

- **Meta Content Library:** We worked through the SOMAR process to access Facebook posts from the Meta Content Library (MCL), extracting posts containing terms matching those used for the Twitter/X API pulls.

- **News Articles:** We acquired headlines and metadata from the GDELT Global Knowledge Graph to track volume and frame of media coverage of H5N1 news articles.

Furthermore, we conducted sampling through keyword filtering and date restrictions tied to USDA confirmed reports of outbreaks in an attempt to allow temporal correlation.

Epidemiological Data Sampling
We acquired ground truth validated H5N1 outbreaks in domestic fowl and wild bird records from the USDA APHIS dataset. These datasets include case confirmation date, site types such as commercial, backyard, zoo etc., species affected, geographic location, and flock size (where available). We will sample these data for the same period as digital discourse data and served as anchor points to synchronize and validate outbreak signals.

Sampling Rationale and Representativeness
We applied relevant sampling in an attempt to secure high-relevance data coinciding with observed outbreak events and stratified temporal sampling to secure adequate coverage in both the times of high outbreak activity and low activity. While our online sources may not be fully representative of the general population, they offer important insights into public awareness, emotional response, and narrative construction in real time. We tried to preserve linguistic diversity where possible.

3.7 Ethical Consideration

Privacy, Security, and Fairness
This study engages with publicly available social media and search data that may include user-generated content with personal information, such as usernames or geolocation metadata. To address privacy concerns, all data will be de-identified, stripping posts of names, handles, or any directly identifying information. Geolocation data will be aggregated to higher-level administrative units (e.g., state or country), with additional suppression protocols (e.g., "<11" display) applied to protect anonymity in low-frequency regions. Furthermore, this study acknowledges the selection bias inherent in social media datasets, particularly those constrained by geography or language. The research team remains aware that digital discourse does not equally represent all communities affected by H5N1.

Harm Mitigation and Responsible Use
Social media users do not explicitly consent to research use of their content. In recognition of this, all data collection and analysis will strictly follow each platform's terms of service and data-use agreements. This includes limiting data collection to public

content and complying with rate limits or content usage policies. The research team will also complete all required training on data use and ethics and adopt responsible data governance practices.

Ethical Communication and Interpretation of Results

Any results produced by the study will be interpreted as exploratory rather than deterministic. The primary goal is to examine whether digital discourse may offer early indicators of outbreak awareness or public anxiety—not to create prescriptive models for surveillance. In communicating results, the research will avoid stigmatizing regions or communities identified with high emotional intensity. IRB-aligned standards regarding vulnerable populations and stigmatization will guide both interpretation and dissemination of findings.

3.8 Trustworthiness of Data

We evaluated and assessed each data source for verifiability, completeness, update frequency, and reproducibility. As part of our data ingestion pipeline, we maintained rigorous provenance and lineage tracking throughout the ingestion and transformation processes.

To further ensure the trustworthiness of our analytical results, we applied rigorous validation techniques tailored to each methodological component:

- Classification Tasks: Accuracy, precision, recall, F1-score, and ROC-AUC.
- Topic Modeling: We use BERTopic to extract thematic structure in the digital discourse, evaluating topic quality with UMass coherence and topic diversity
- A time-aware logistic regression model with lead–lag stigma features to assess whether digital emotional signals precede official outbreak announcements.

Model performance was validated using manually labeled subsets. A benchmark F1-score > 0.75 was used as a threshold for deployability in exploratory modeling.

In addition, we utilized SHAP analysis (SHapley Additive exPlanations) to evaluate sentiment scoring prediction and identify which words or phrases contributed to a sentiment prediction. For example, Example: For a social media post classified as negative sentiment, SHAP might highlight words like “panic”, “contagion”, “no cure” with a higher feature importance for the context. This method provided us with further explainability in our sentiment prediction while paving the way to provide additional transparency and trustworthiness to the analytical pipeline.

3.9 Assumptions, Delimitations, and Limitations

This study is based on several key assumptions. First, it assumes that the digital content collected from platforms such as Twitter and Facebook reliably reflects public sentiment and awareness regarding H5N1. It is also assumed that the timestamps, locations, and metadata associated with these digital traces are sufficiently accurate to allow for meaningful temporal and regional analysis.

In terms of delimitations, the study focuses only on English-language content due to the availability of pretrained NLP models and tools. It also restricts its analysis to specific time frames surrounding known H5N1 outbreaks, rather than performing continuous surveillance. Furthermore, the study primarily analyzes publicly available data, excluding private or subscription-based sources such as proprietary news feeds.

The study also faces certain limitations. Digital signals are inherently biased by platform usage patterns and demographic differences; for instance, Twitter users may not represent the general population. Sentiment analysis models, especially those trained on generic corpora, may misinterpret context-specific language, sarcasm, or non-English idioms. Additionally, the absence of real-time ground-truth data from health departments may limit the precision of outbreak correlation or prediction. Despite these constraints, the study seeks to generate meaningful insights into the dynamics of digital discourse surrounding H5N1 and its potential role in public health monitoring.

Chapter 4: Analysis

4.1 Overview of the Early-Warning Modeling Framework

The stigma labeling of the digital signals acquired from the Meta Content Library helped us convert raw social media text into daily emotional signals such as fear and urgency. These signals serve as behavioral indicators that can rise weeks before official outbreak announcements. By using keyword rules and sentiment scoring, the system produces interpretable and scalable measurements of public emotional response. In this section, we describe how we turn raw social media posts into daily "stigma signals" and then use those signals to build a simple early-warning model for H5N1 outbreaks.

These stigma features form the foundation of the time-aware early-warning model, and they show promising signs of predictive power even with simple logistic regression, suggesting a real behavioral lead time in digital discourse before outbreaks are formally detected. The core idea is straightforward: people often talk about a disease before official systems record an outbreak. By labeling posts for emotions such as fear or urgency and then aggregating those labels over time and by country, we obtain behavioral indicators that may start to change days or weeks ahead of formal reports.

Using a combination of keyword-based rules and sentiment scoring, we construct daily measures that capture the emotional tone of online conversations. Even with a relatively simple logistic regression model, these features show encouraging signs of predictive power, suggesting a real behavioral lead time in digital discourse.

Digital "stigma signals" such as fear, urgency, humor, and neutrality serve as behavioral indicators that may precede outbreak announcements. The labeling process transforms raw text into interpretable emotional categories suitable for early-warning modeling.

This analysis covers:

- Why stigma labeling matters
- Stigma classification methodology
- Pipeline steps with examples
- Strengths and weaknesses of rule-based labeling
- Visualization of the stigma classification process
- Recommendations for improvement

4.2 Purpose of Stigma Labeling

Regarding public health events, digital populations often express:

Fear	Anxiety about disease spread
Urgency	Calls to act, alerts, rapid information sharing
Blame	Directed anger or distrust (merged into fear)
Humor	Trivialization or dismissal
Neutral	Unrelated or ambiguous content

These emotional signals can reveal collective behavioral amplification, an early indicator of evolving outbreaks. This is consistent with the Social Amplification of Risk Framework (Kasperson et al., 1988). Our stigma labeling approach operationalizes these emotional categories into quantifiable, daily, country-level features.

4.3 Stigma Classification Pipeline

4.3.1 Data Sources

This study integrates digital behavioral signals with multi-source epidemiological intelligence to evaluate whether shifts in online emotional and semantic expression precede confirmed H5N1 outbreaks. The methodological pipeline consists of preprocessing and labeling social-media text, extracting sentiment and topic-based semantic structure, aggregating behavioral signals into daily country-level indicators, harmonizing outbreak data from multiple reporting systems, generating retrospective early-warning labels, and constructing time-series features to support early-warning modeling. All data streams were aligned at the *country–day* level, enabling joint temporal analysis of emotional, semantic, and epidemiological dynamics leading to outbreak events.

4.3.2 Text Preprocessing and Normalization

In order to conduct a meaningful analysis of Facebook posts, it was necessary to ensure that the content was reasonably comparable. Raw social media text is noisy; users paste links, type in all caps, switch languages mid-sentence, use emoji instead of words, and generally break most of the assumptions underlying standard NLP tools. To address this, we started with a fairly pragmatic preprocessing pipeline designed to reduce obvious noise without over-engineering.

First, we removed URLs, emoji, and obvious boilerplate characters (for example, long strings of punctuation or repeated symbols). Posts like "BREAKING!!!!, or https://..." are very common, and without this step those elements dominate the token space but contribute almost nothing to the emotional signal we care about. We also stripped extraneous whitespace, including multiple spaces, line breaks in odd places, and so on in an attempt to process the texts more consistently.

Next, we applied case normalization, converting everything to lowercase. This is a standard move, but it is worth stating: people often use capitalization for emphasis ("THIS IS BAD"), and we lose that nuance here. We decided that the gain in vocabulary consistency outweighed the loss in emphasis. In a later, more refined version of the pipeline, one could track "all caps" as a separate signal, but we did not do that in this first pass.

We then tokenized the posts into word units. In practice, this means splitting on spaces and standard delimiters after cleaning. We did not use a highly sophisticated tokenizer. No sub-word models or language-specific tokenization rules were used because our primary goal at this stage was to support lexicon matching and sentiment analysis, not fine-grained syntactic parsing. This is a trade-off: some edge cases (compound words, hashtags with attached text) are not handled perfectly, but the approach is simple and transparent enough to debug.

A small but important design decision was that we did not remove semantic content such as disease names, locations, or domain-relevant keywords. In many generic NLP pipelines, one might strip out stopwords or extremely frequent tokens to reduce dimensionality, but here those "boring" words sometimes carry context that interacts with emotional language. For instance, "outbreak in chickens" and "outbreak in humans" both include common words, yet they sit in very different risk contexts. To avoid accidentally throwing away this nuance, we chose not to apply aggressive stopword removal.

In short, the preprocessing steps included:

- Removal of URLs, emojis, and obvious boilerplate symbols
- Removal of extraneous whitespace and normalization of spacing
- Case normalization to lowercase
- Basic tokenization into word units

These relatively simple operations already went a long way toward making the posts more uniform across different writing styles. They also made subsequent lexical and sentiment-based classification more stable, because the same word is more likely to look the same across posts (instead of appearing in multiple noisy variants). The pipeline is not perfect, and it certainly misses some stylistic nuance. However, it provides a clean enough foundation for the stigma labeling and early-warning modeling that follow.

4.3.3 Stigma Classification of Digital Text

Once the posts were cleaned and tokenized, we needed a way to characterize the emotional posture users were taking toward disease-related content. This proved to be a delicate step. Social-media posts are short, often ambiguous, and people don't always state their emotions explicitly. However, certain linguistic cues tend to show up consistently when individuals express fear, urgency, or even humor about a public health event. Our goal was to design a system that could reasonably sort posts into broad "stigma" categories based on these recurring patterns.

To achieve this, we relied on a combination of curated lexicons and a small set of rule-based decisions. Rather than building a heavy-weight classifier trained on labeled data, which was not available to us at scale, we opted for a transparent approach that we could inspect and adjust. We started by developing lexicons for four main emotion categories:

- Fear: words and phrases signaling a perceived threat or danger, or simply anxiety about the situation. In our dataset, these ranged from very direct terms ("scary," "dangerous") to subtler cues that still conveyed unease.
- Urgency: expressions that suggest people feel something requires immediate attention. These can be alarms ("act now"), warnings, or even rushed sharing ("urgent update").
- Humor: joking, sarcasm, or dismissive comments. Social media users frequently turn serious events into material for memes or casual banter, and these posts needed to be distinguished as they behave quite differently from genuine concern.
- Blame: posts expressing anger or frustration toward institutions or groups. This was later merged with the Fear category, because blame often co-occurs with heightened negative affect and tends to follow similar temporal patterns during outbreaks.

This lexicon-driven approach only works if the system can decide what to do when multiple cues appear at once; or, more commonly, when nothing obvious appears at all. To handle this, we created a priority order. Humor was evaluated first, because it tends to stand out lexically and can overshadow other emotional signals. Next came blame, then urgency, and finally fear. The idea here was simply to reduce ambiguity: if someone is making a joke, the humorous tone usually dominates the post regardless of other keywords.

For posts that didn't match any of our lexicon categories, which occurred more frequently than expected, we relied on VADER. It is a lightweight sentiment model commonly used for short, informal text. VADER doesn't understand disease risk, but it does capture broad emotional polarity. When the model produced a strongly negative score, we treated that as a sign of fear or distress and labeled the post accordingly. Neutral or weakly valanced posts were included in the neutral category.

Finally, we encoded the resulting labels using a one-hot representation across the four dimensions (fear, urgency, humor, and neutral). This made downstream aggregation much more straightforward: we could simply count how many posts in a given country and day fell into each emotional bucket, turning messy human expressions into structured signals that could be tracked over time.

Overall, the classification process is admittedly imperfect. Natural language is messy, and emotional nuance cannot always be captured by a few rules. However, it gives us a workable, transparent way to approximate how public sentiment is shifting as disease-related events unfold. For early-warning modeling, we believe this consistency is more important than capturing every subtlety.

4.3.4 Sentiment Polarity

In addition to the discrete stigma labels, we also assigned each post a continuous sentiment polarity score. The reason behind this was that the stigma categories tell us *what kind* of emotion a post expresses (fear, urgency, humor, neutral), but they do not capture *how strongly* that emotion is felt. For example, two posts might be labeled as fear with one reading like mild concern and the other like outright panic. We wanted a way to quantify that difference.

To do this, we applied a standard sentiment tool that produces a single polarity score for each post, ranging from strongly negative through neutral to strongly positive. In practical terms, we ran it on the preprocessed text and stored the resulting "compound" score as an extra column in the data. We did not treat this score as an absolute ground truth about emotion. Instead, we used it as a rough but useful measure of emotional intensity. This tells us whether the general tone of a country's posts was drifting more towards the negative or staying fairly calm.

Polarity played three roles in the analysis. First, it gave us a way to check our stigma labels. If a large cluster of posts tagged as fear consistently showed neutral or even positive polarity, that was a signal that our lexicons might be overfiring on certain technical phrases or institutional language. Likewise, if humor-labeled posts were overwhelmingly negative, they were inspected more closely to distinguish between dark sarcasm and actual misclassification. This cross-checking helped us avoid leaning too heavily on the rule-based labels without any external reference.

Second, polarity allowed us to summarize the overall emotional climate at the country–day level. When we aggregated posts by country and date, we not only counted how many posts expressed fear or urgency, but also computed the mean polarity and the share of strongly negative posts for that day. These aggregated polarity measures became part of the early-warning feature set. For instance, they helped us distinguish between a country that had a moderate amount of fear-related language but relatively mild sentiment, and a country where the same share of fear posts was accompanied by very sharp negativity.

Third, we treated polarity as an independent time-varying signal. Similarly to building lags and rolling averages for the stigma proportions, we applied the same to mean polarity. This allowed the model to detect gradual shifts in tone, even when the raw proportion of specific emotions remained relatively stable. In our experience, these slow drifts in negativity often carried more information than a single dramatic spike.

Overall, sentiment polarity did not replace the stigma labels; it sat alongside them. The labels gave us a coarse map of which emotional "mode" a post belonged to, while polarity added shading and depth. Together, they provided a more nuanced view of how people were reacting over time, which proved valuable in identifying the days that quietly precede an outbreak.

4.4 Daily Country-Level Stigma Aggregation

Once each post had been assigned to a stigma category, we needed a way to make sense of these labels at the population level. A single post, by itself, rarely tells us much. People use social media for all sorts of reasons unrelated to epidemiological events. For example, venting frustrations, sharing jokes, reposting news headlines without much thought, or reacting emotionally to events that have nothing to do with disease. But when we begin to look at the distribution of these emotions over time, especially when narrowed to a specific country, a clearer picture starts to emerge. It is these collective shifts, rather than isolated posts, that hold potential value for early-warning systems.

To capture this collective dynamic, we aggregated all labeled posts for each country and day. We could not rely on the raw number of posts showing fear or urgency as they varied wildly between countries with large online populations and those with relatively small ones. Instead, we computed proportions, which produced a more balanced and interpretable picture of emotional trends. For any country ‘c’ on date ‘t’, the daily fear signal was calculated as:

$$\text{stigma\_fear}_{c,t} = \frac{\#(\text{fear posts}_{c,t})}{\#(\text{all posts}_{c,t})}$$

The same logic was applied to urgency, humor, and neutrality.

The choice to use proportions may seem like an insignificant technical detail, but it underpins the entire analytical framework. Social media activity is uneven by nature. Countries with large populations or high connectivity produce far more posts than smaller or less-connected regions. Had we used raw counts, the emotional behavior of people in large countries would dominate our analyses simply by volume, rather than by any meaningful signal. By normalizing within each country-day, we ensure that a small but significant shift in fear in a lower-volume country is treated with the same conceptual weight as a larger shift occurring in a much busier digital environment.

There is also the issue of fluctuating activity within countries. Social media volumes can spike or dip for reasons that have nothing to do with disease. Weekends tend to look different than weekdays, political events draw bursts of attention, and cultural holidays can reduce posting activity altogether. Using proportions helps smooth over these structural irregularities: what matters is not how much people are posting, but what they are expressing in the posts.

In practice, these daily stigma proportions began to reveal subtle but meaningful patterns. Fear tended to rise slowly and steadily rather than in sudden jumps. Urgency, on the other hand, often showed brief spikes that might correspond to sudden bursts of news coverage or rumor circulation. Humor behaved differently, sometimes increasing during early phases of discussion when an outbreak was not yet taken seriously, and then tapering off as the situation gained gravity. These patterns were not uniform across countries, which highlighted differences in social media cultures; people express emotions in different ways depending on linguistic, cultural, and informational contexts.

Another benefit of this approach in aggregating stigma labels is that it allowed us to align emotional signals with epidemiological data on a temporal grid. Without aggregation, the dataset would remain a noisy stream of individual posts, difficult to match against outbreak timelines or to feed into any predictive model. Once turned into daily proportions, however, the same data can be treated like a structured time series. This is essential for early-warning systems, which rely on the idea that signals unfold over time and that the past can be used to infer something about the near future.

Of course, aggregation comes with trade-offs. By condensing thousands of posts into a single daily score, some nuance is inevitably lost. A sudden burst of fear expressed in just a few highly influential posts may not appear as strongly once averaged across the day. The approach also assumes that all posts carry roughly equal weight, which is not always true as some users have far larger audiences or different levels of credibility. Nonetheless, for a first-pass, large-scale model that aims for transparency and simplicity, this method offers a reasonable compromise between granularity and interpretability.

In short, daily country-level stigma aggregation provides a stable, comparable way of tracking how collective emotional behavior shifts over time. These processed signals act as the backbone of our early-warning framework, enabling us to examine whether emotional changes in digital discourse precede officially recognized outbreak events. Without this aggregation step, the underlying noise of social media would overwhelm the very patterns we are trying to detect.

4.5 Topic Modeling of Digital Discourse

To make sense of what people were talking about, beyond whether posts were fearful, urgent, or humorous, we applied topic modeling to the user-generated content from the Meta Content Library and Twitter/X. The aim was not to discover a large number of very fine-grained topics, but to recover a small set of recurring narratives that show up across countries and over time, and then relate those narratives to our stigma and early-warning signals.

We restricted this step to English-language posts and used the same cleaned text that fed into the stigma and sentiment pipeline. Very short or malformed posts were dropped, as they tended to behave like noise in the topic model and did not meaningfully contribute to any theme.

4.5.1 TF–IDF Representation and NMF Model

We first converted the corpus into a standard TF–IDF representation. Each post was tokenized and mapped into a high-dimensional vector space using TfidfVectorizer with English stop words and a capped vocabulary (5,000 terms). This representation gives higher weight to words that are relatively distinctive for a given post, while down-weighting very common terms that appear often.

On top of this TF–IDF matrix, we fit a Non-negative Matrix Factorization (NMF) model with ten components. In practice, this means approximating the document–term matrix as the product of:

- a document–topic matrix (how strongly each post loads on each topic), and
- a topic–term matrix (how strongly each word contributes to each topic).

We chose ten topics as a compromise between interpretability and coverage; fewer topics tended to lump together several distinct narratives, while more topics produced redundant or unstable clusters. The model was initialized with the standard non-negative SVD scheme and run with a fixed random seed so that the decomposition was reproducible.

For each post, we took the topic with the highest weight in its document–topic vector as its primary topic assignment. This gave us a simple mapping from posts to topics that could be merged back with countries, dates, and stigma labels.

4.5.2 Topic Interpretation and Labeling

Once the NMF model converged, we used the topic–term matrix to make sense of what each factor represented. For every topic, we pulled out the highest-weighted terms (e.g., "birds, wild, dead, avian, sick, poultry, virus" for Topic 0 or "human, case, flu, reported, health, death, patient" for Topic 2) and treated these as candidate keywords. We then read through a sample of posts that loaded strongly on each topic to check that the words and the actual content lined up in a way that made sense.

This iterative back-and-forth, looking at top words and then reading real posts, led us to a set of qualitative labels that we use throughout the paper instead of raw topic IDs. In particular:

- **Topic 0 – Wild bird die-offs & spillover risk** (birds, wild, dead, avian, sick, poultry, virus, contact)
- **Topic 1 – Government alerts & biosecurity (UK)** (premises, gov, information, available, UK, confirmed, biosecurity, keepers, follow)
- **Topic 2 – Human H5N1 cases & outcomes** (human, case, flu, reported, health, death, patient)
- **Topic 3 – Pet food recalls & companion animals** (pet, food, recall, cat, pets, warning, raw)
- **Topic 4 – Dairy and cattle H5N1 infections** (dairy, milk, cows, cattle, herds, raw)
- **Topic 5 – Pandemic framing, COVID & vaccination** (pandemic, covid, virus, vaccine, humans)

- **Topic 6 – Public risk communication & safety** (reports, public, risk, guidelines, safety, health)
- **Topic 7 – General H5N1 outbreak discussion** (flu, bird, virus, avian, humans, spread, outbreak)
- **Topic 8 – Technical HPAI surveillance & confirmations** (influenza, highly pathogenic avian influenza, HPAI, confirmed, county, strain)
- **Topic 9 – Poultry farm outbreaks & agriculture** (outbreak, poultry, farm, province, agriculture)

These labels are necessarily interpretive to an extent, but they capture the dominant narratives that occurred repeatedly in the data: wild bird die-offs and spillover concerns, formal government alerts and biosecurity advice, human case reports and clinical outcomes, pet and companion-animal exposure via recalled food, spread into dairy and cattle systems, pandemic-style framing and vaccine debates, more general public risk communication, broad outbreak chatter, technical HPAI surveillance language, and poultry-farm outbreaks with agricultural impacts.

Using these descriptive labels rather than numeric topics alone made it easier to describe how digital discourse evolved. For example, to say that fear increased specifically within “Human H5N1 cases & outcomes” and “Dairy and cattle infections,” rather than simply reporting that “Topic 2” and “Topic 4” went up.

4.5.3 Visualization of Topic Structure

To make the topics more tangible, we used two complementary visualizations. First, we created a grid of topic panels (Figure 4). For each topic, we plotted the top eight TF–IDF-weighted terms as a horizontal bar chart and used the qualitative label as the panel title (e.g., “Human H5N1 cases and clinical outcomes”). This figure lets the reader see, at a glance, what each topic is about, without needing to parse the full vocabulary. It also makes clear that the model is picking up distinct storylines such as pet food recalls, dairy herd infections, pandemic framing rather than simply rediscovering “avian flu” repeatedly.

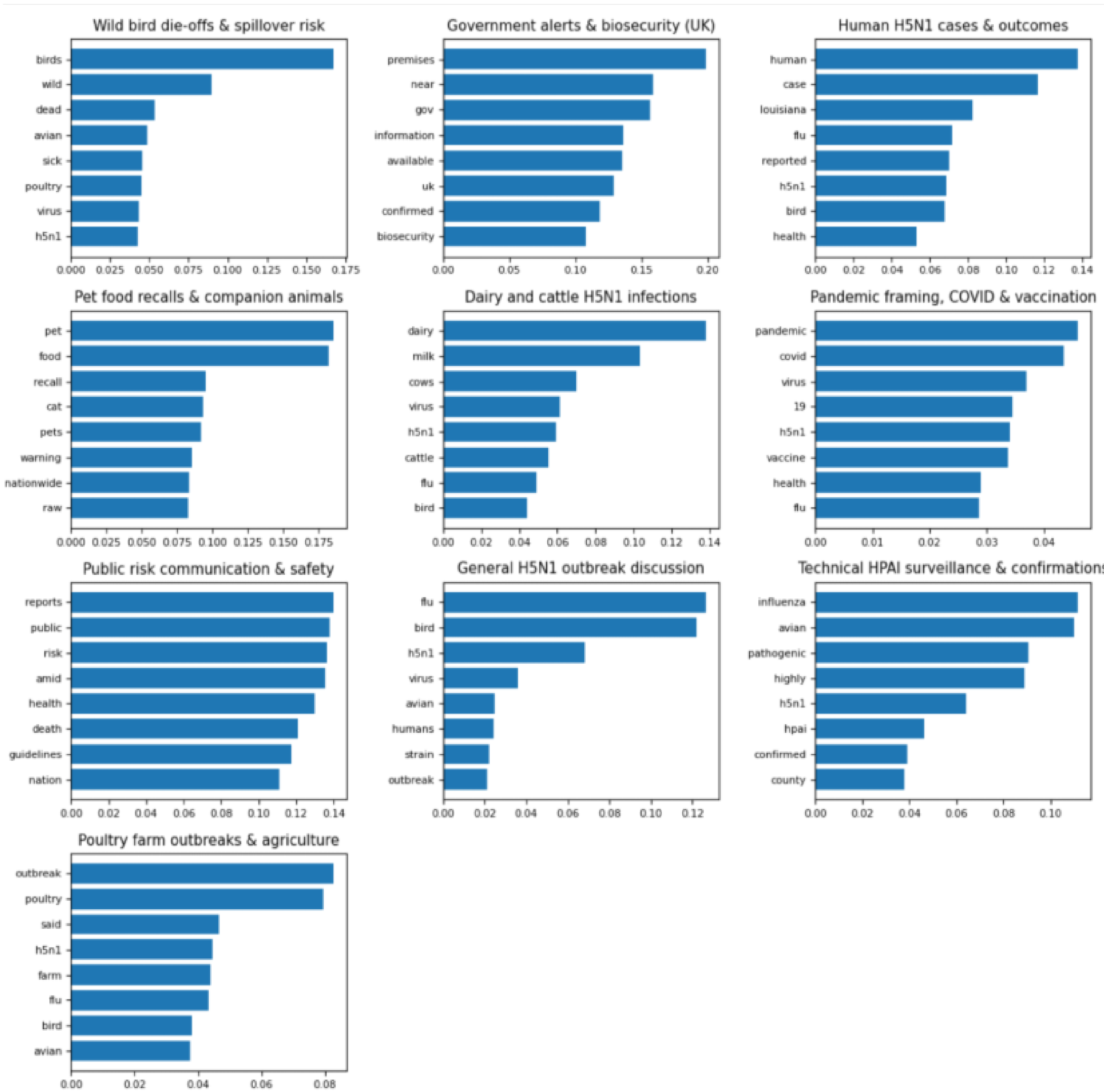


Figure 4: Grid of topic panels

Second, we constructed time-series visualizations of topic prevalence (Figure 5). After assigning each post to a primary topic, we aggregated to daily country–topic counts and then converted these to topic shares within each country-day. For clarity, we focused on a small set of key topics (e.g., human cases, dairy and cattle infections, pandemic framing) and plotted their 7-day smoothed volumes over time. These plots allowed us to see when certain narratives “lit up” in the digital discourse and how they co-evolved with stigma signals and official outbreak reports.

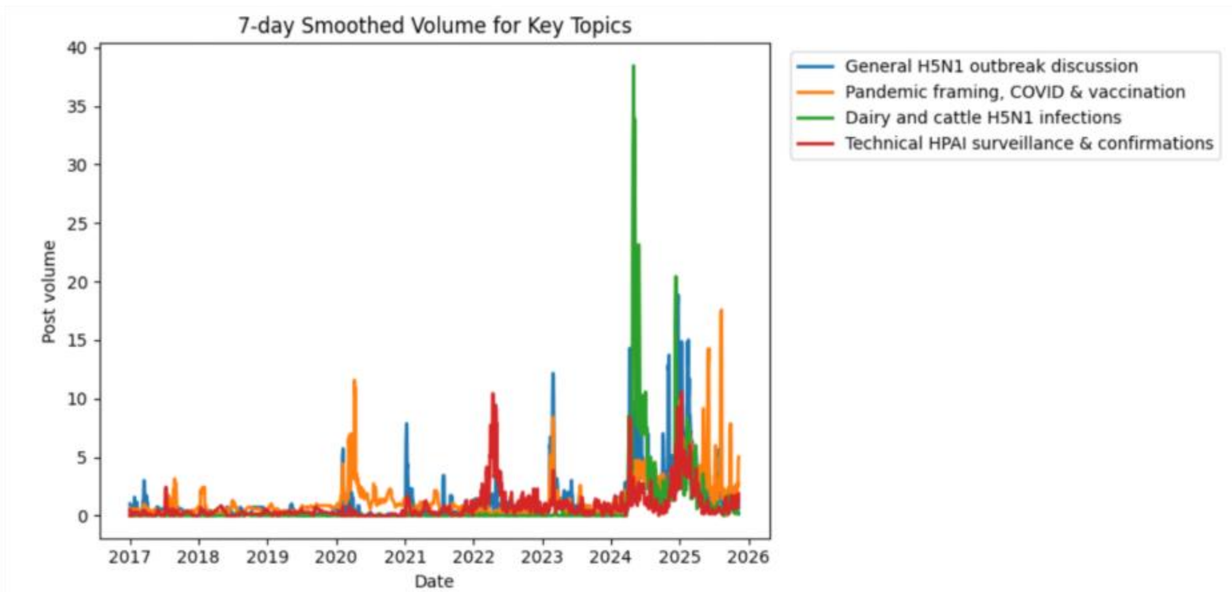


Figure 5: Time-series graph

Together, the topic labels and visualizations served two purposes. On the methods side, they provided a sanity check that the NMF model is capturing meaningful themes. On the analytical side, they enabled us to make more informative statements in interpreting the early-warning model. For instance, rather than stating that “fear increased,” we were able to infer that fear increased specifically within narratives about human cases or cattle infections.

4.5.4 Sentiment Polarity Overlay on Topic Model

To make the topics slightly less abstract, we overlaid sentiment polarity on the ten labeled themes and summarized the result in a stacked bar plot (Figure 6). For each topic, we took the VADER compound score for every post, bucketed it into negative, neutral, or positive, and then computed the distribution of these three categories within that topic. The bars in the figure each sum to 1.0, and the plots show the *share of posts* in each topic that fall into each sentiment bucket.

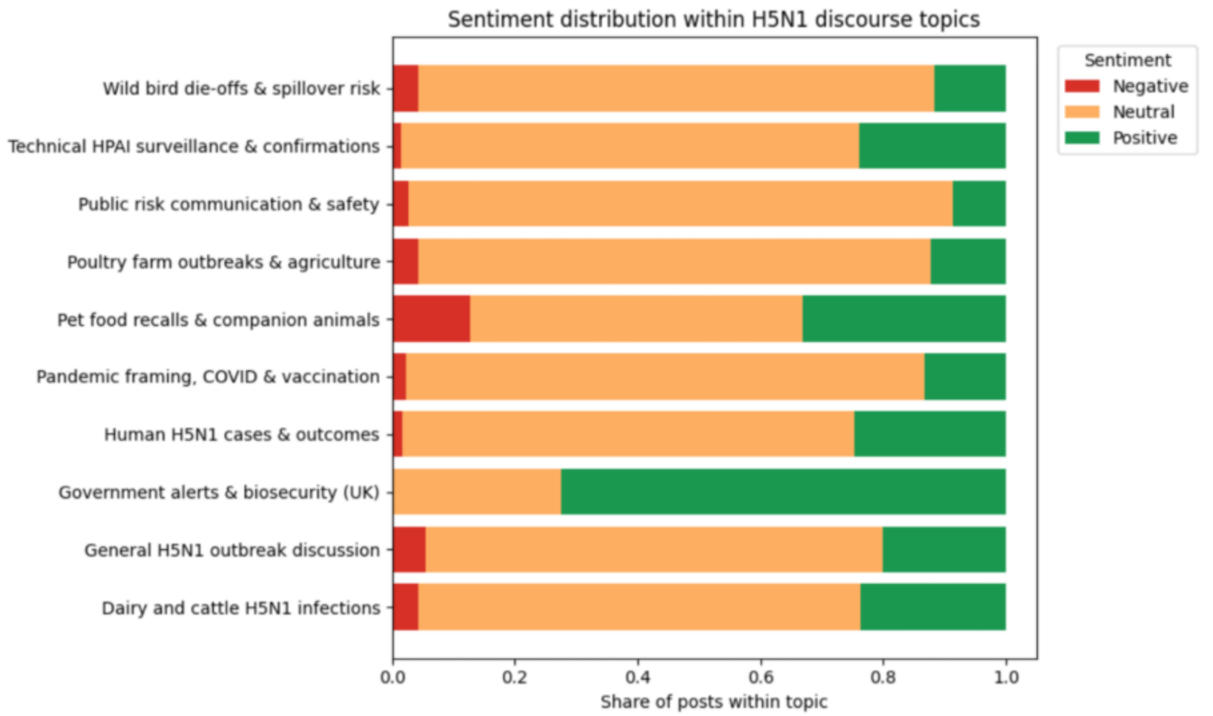


Figure 6: Sentiment distribution bar chart

Several patterns emerge from the analysis of the graph:

- Most topics are dominated by neutral or mildly positive language. Themes such as "Technical HPAI surveillance & confirmations", "Public risk communication & safety", and "Dairy and cattle H5N1 infections" are largely informational. They carry relatively little overt negativity and instead read like routine reporting or guidance.
- "Government alerts & biosecurity (UK)" is almost entirely neutral and positive. These posts tend to stress what officials are doing and what keepers should do next, which naturally produces reassuring or matter-of-fact phrasing rather than panic.
- A small number of topics show a visibly higher negative component. In particular, "Pet food recalls & companion animals", "Wild bird die-offs & spillover risk", "Poultry farm outbreaks & agriculture", and the more catch-all "General H5N1 outbreak discussion" all include a non-trivial slice of negative posts. These are the storylines where people are more likely to express worry, anger, or frustrations about sick pets, dead wild birds, culling, or the possibility of "another pandemic."
- Even for emotionally charged topics like "Human H5N1 cases & outcomes" and "Pandemic framing, COVID & vaccination", the figure shows that a large fraction of posts remain neutral in tone. Much of the human-case and pandemic talk is framed

through news headlines, official updates, or contextual explanations rather than personal fear narratives.

This pattern is useful when we think about stigma. If we only considered topic prevalence, we might conclude that any rise in topics such as human-case or cattle-infection signals mounting public anxiety. The sentiment overlay adds more nuance; it shows that some topics (especially government alerts and surveillance) can grow in volume without much change in emotional tone, while others, like pet food recalls or general outbreak chatter, carry a heavier negative component even when their overall volume is modest. In the modeling stage, this motivated us to treat topic membership, sentiment polarity, and stigma labels as complementary signals:

- topic labels explain *what* the conversation is about;
- sentiment shows *how* people are reacting within that topic on a negative–positive axis; and
- stigma labels (fear, urgency, humor, neutral) capture more specific emotional postures that may not align perfectly with generic negativity.

The sentiment-over-topics view reassures us that the model is not simply keying off generalized negativity. Instead, it helps isolate those topic–sentiment combinations such as negative posts about human cases, livestock infections, or spillover in wild birds that are more plausibly linked to perceived risk and, ultimately, to early-warning behavior.

These topics and sentiment associations align well with the principles of SARF as outlined in Chapter 2. According to this framework, risk narratives that carry symbolic or emotionally charged elements intensify public perceptions of threat by reinforcing emotions such as fear and anxiety, and by fostering stigma. The strong association observed between narratives involving wild bird die-offs, potential contamination of companion animals, and reported human infections, and the accompanying signals of fear and stigma, indicates that these amplification mechanisms are actively operating within digital discourse.

4.6 Construction of Epidemiological Ground Truth

4.6.1 Sources of Data

One of the foundational steps in building an early-warning system is determining what qualifies as an outbreak. Different organizations report outbreaks for different reasons, on different timelines, and often with different levels of confirmation. Some systems pick up weak early signals, such as rumor-level reporting or news chatter, while others only document events after laboratory confirmation or official verification. No single source, on its own, gives a complete or perfectly reliable picture. To address this, we compiled outbreak indicators from four independent sources, each contributing a different perspective:

- GDELT: global news-derived event signals capturing early journalistic reporting.
- ProMED: expert-curated alerts on human and animal disease events, often appearing before formal institution-level announcements.
- FAO EMPRES-i: structured animal and domestic flock outbreak data, offering detail on agricultural and veterinary incidents.
- WHO Disease Outbreak News: formally verified reports of confirmed human cases.

Together, these sources form our "information ecosystem" for outbreak detection. Some signals surface quickly but may be noisy; others arrive more slowly but are more authoritative. By integrating all four, we aimed to produce a ground truth that is both sensitive and robust.

$$\text{fused\_outbreak}_{c,t} = \begin{cases} 1, & \text{if any source reports an outbreak on day } t \\ 0, & \text{otherwise.} \end{cases}$$

This fusion rule is deliberately simple. A day was treated as part of the outbreak timeline if any single system identified an outbreak-related event. In other words, we prioritized sensitivity over strict verification. For the purpose of early-warning studies, minimizing missed events is a higher priority than eliminating noisy signals.

We rely on multiple sources because each reporting system has its own strengths and weaknesses. GDELT, for instance, may capture early speculation or breaking news sooner than official channels, but it can also reflect media hype. WHO reports are highly reliable but typically appear later in the outbreak cycle. ProMED offers expert judgment and often detects unusual patterns, while FAO EMPRES-i provides precision for animal outbreaks that may precede human cases.

By combining them, we mitigate the inherent weaknesses of any one source. This blended approach helps reduce missingness, smooths out reporting delays, and produces a more complete outbreak timeline that is better suited for evaluating whether digital behavioral signals truly precede recognized events.

4.6.2 Interpretation and Practical Implications

Using a harmonized outbreak indicator changes how we interpret the relationship between digital emotion signals and real-world events. A rise in fear or urgency may not correspond neatly to WHO-verified cases alone, but it might precede early media reports or veterinary detections. In this sense, the fused outbreak series reflects a more realistic progression of information rather than a single institution's timeline.

This broader perspective is crucial when evaluating whether digital signals can serve as early-warning indicators. If we match digital behavior only to the latest, most formal reports, we risk undervaluing the predictive power of online discourse. By grounding the analysis in a multi-source outbreak timeline, we gain a better understanding of how public attention aligns with the earliest available hints of disease activity.

4.6.3 Construction of Early-Warning Labels

Identifying when an outbreak begins is one challenge; determining when early-warning signals should be expected is another. To evaluate whether changes in digital stigma indicators can serve as precursors to real-world events, we needed a way to mark the days leading up to an outbreak as "early-warning" periods. This labeling is central to the entire modeling task, because the model is not trying to predict the outbreak day itself but rather trying to recognize the behavioral lead-up to it.

We approached this by defining a retrospective temporal window. For each country, once an outbreak day D was identified (based on the fused ground-truth timeline described earlier), we marked the 30 preceding days, from D -30 through D -1, as belonging to the early-warning period. All other days were given a label of 0.

Formally, the early-warning label was defined as:

$$\text{early\_warning}_{c,t} = \begin{cases} 1, & t \in [D-30, D-1] \\ 0, & \text{otherwise.} \end{cases}$$

Although this rule appears simple, several layers of reasoning behind it for optimum efficacy.

A 30-day lead time reflects a balance between overly conservative and overly generous definitions of what counts as "early." Many public-health reporting systems operate with substantial lags, sometimes days, sometimes weeks, between real-world events and formal recognition. During that time, public attention, rumor circulation, and emotional reactions often begin to intensify. If the window is too short (e.g., only a few days), we risk missing this anticipatory period entirely. If it is too long (e.g., 60–90 days), we begin to capture unrelated variations in social media behavior that has nothing to do with outbreak dynamics.

A crucial aspect of this framework is that we treat the days before an outbreak as the actual target for prediction. Outbreaks are often accompanied by immediate spikes in attention, media coverage, and emotional discourse. Predicting those spikes is less interesting, since they typically reflect events that have already become visible to authorities. Instead, our goal is to understand whether online emotional dynamics begin shifting subtly *before anyone formally announces* that something is happening. For example:

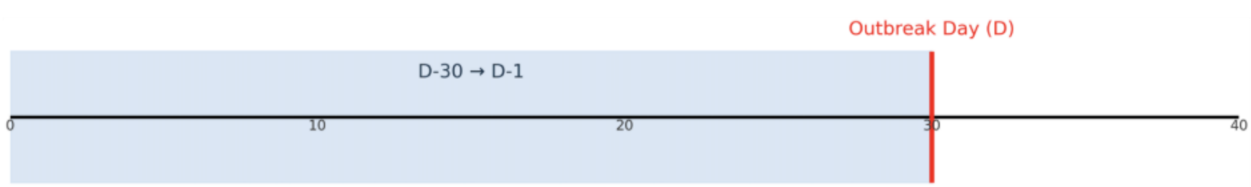
- Fear may gradually rise as people witness unusual events in their communities.
- Urgency may spike as rumors or unconfirmed reports circulate.
- Humor or dismissiveness may decline as risk is perceived to be more serious.

By labeling these pre-event days as 1 (early-warning) and the rest as 0, we allow the model to focus on the behavioral buildup rather than the event peak.

It's worth noting that these labels are created retrospectively, after we know when an outbreak officially occurred. In a real deployment, the model would not know the future; it would need to infer risk from emotional signals alone. However, retrospective labeling is standard practice in early-warning research because:

1. It creates a clear and consistent training target.
2. It avoids circular reasoning (e.g., labeling days based on emotional data alone).
3. It ensures comparability across countries with very different outbreak timelines.

This backward-looking approach essentially asks:
If we imagine ourselves on day t, and we don't yet know that an outbreak will occur on day D, do the emotional signals on day t resemble those that, historically, tended to precede outbreaks?



30-day early-warning window

Some countries experienced more than one outbreak event during the study period. In these cases, we applied the same logic repeatedly: each outbreak contributed to its own

30-day pre-event window. These windows were allowed to overlap, which occasionally produced stretches where large portions of the calendar were labeled as early-warning. This is not a flaw. It reflects the reality that some regions experience clusters of outbreaks rather than isolated events.

4.6.4 Trade-Offs and Limitations of Labeling Strategy

Like any operational definition, this labeling scheme has limitations:

- **Sensitivity to the chosen window size:**
  A 30-day window captures medium-range behavioral shifts but may miss very early or very late signals.
- **Uniformity across countries:**
  A 30-day period may not be equally appropriate for small versus large countries, or for capturing outbreaks of varying severity.
- **Dependence on outbreak detection quality:**
  If formal reports are delayed or inconsistent, the “true” early-warning period may be misaligned.

Despite these shortcomings, the approach provides a transparent and interpretable starting point for detecting temporal associations. More complex schemes such as adaptive windows or multi-level temporal labeling may be valuable for future work, but the 30-day window strikes a reasonable balance for this first model.

4.6.5 Time-Series Feature Engineering

Daily stigma proportions on their own tell us something about the emotional “snapshot” for a given country and date. However, outbreaks unfold over time, and so do people’s reactions to them. If we only look at a single day at a time, we risk missing the gradual build-up or decay of fear, urgency, or humor that might matter most for early-warning purposes. To capture this temporal structure, we built a small set of lagged and smoothed features from the raw daily signals.

Lagged features: looking back in time

The first idea is straightforward: what happens today may depend on what people were feeling a few days ago. To reflect on this, we created lagged versions of the stigma signals. For fear, a 3-day lag is defined as:

$$\text{fear\_lag3}_{c,t} = \text{stigma\_fear}_{c,t-3}.$$

We constructed 1-, 3-, and 7-day lags for fear, and analogously for urgency and humor. These lags give the model access to short- and medium-term emotional history:

- A 1-day lag captures very recent shifts ("What did fear look like yesterday?").
- A 3-day lag reflects slightly slower dynamics, such as the impact of a weekend news cycle or a cluster of alarming posts.
- A 7-day lag starts to pick up weekly patterns, smoothing over day-to-day noise while still being close enough to the outbreak to be relevant.

In practical terms, this means that when the model makes a prediction for a given day, it does not only see today's level of fear or urgency; it also sees how those levels have evolved over the past week.

Rolling averages: emotional "momentum"

While lagged values tell us what the signal looked like on specific past days, we also wanted to capture a sense of momentum. For instance, we wanted our model to detect whether concern has been consistently high or low over a period, rather than spiking briefly. To capture this, we computed rolling averages.

For fear, a 7-day rolling average is given by:

$$\text{fear\_roll7}_{c,t} = \frac{1}{7}\sum_{i=0}^{6} \text{stigma\_fear}_{c,t-i}.$$

This simple formula averages the current day and the previous six days. The result is less sensitive to one-off jumps and more reflective of sustained emotional states. For instance, a single news story might produce a sharp one-day burst of fear that quickly dies down; a high rolling average, in contrast, suggests that people have been worried for several days in a row.

We computed similar rolling averages for urgency and humor. In the context of early-warning, persistent urgency might indicate ongoing public concern or repeated calls to action, while a sustained drop in humor could signal a shift from joking about an outbreak to treating it more seriously.

4.6.6 Feature Significance and Trade-offs

These engineered features that represent lags and rolling averages give the model a richer description of how emotions are changing, not just where they are on a single day.

Conceptually:

- Lag terms help the model recognize patterns like “fear was low last week but suddenly jumped three days ago, and is still high today.”
- Rolling averages help tease apart fleeting spikes from more durable shifts in public sentiment.

We deliberately kept the feature set compact and interpretable rather than throwing every possible transformation at the problem. This makes it easier to identify what the model is actually learning later on, especially when we examine SHAP values and try to explain why certain days are flagged as high early-warning risk.

As with any feature engineering, there are trade-offs. These transformations assume that relatively short temporal windows (one to seven days) are the most informative for early-warning, which may not always hold. Some outbreaks might have much slower build-up periods, or emotional reactions could lag behind official reporting rather than precede it. Nonetheless, our initial analyses suggested that these modest, time-aware features capture enough structure to improve prediction performance without making the model excessively complex.

These time series design elements align with the theoretical background in digital epidemiology. Prior research indicates that cumulative emotional or behavioral patterns offer more reliable reflections of disease dynamics than isolated one day fluctuations. The incorporation of lagged features and rolling averages for fear and urgency in this study provides a structured means of capturing these cumulative changes in public sentiment. This approach demonstrates that sustained emotional trajectories can provide more robust early warning indicators than daily volatility alone.

In summary, the time-series features translate daily stigma signals into dynamic emotional trajectories, such as patterns of rising, falling, or sustained concern, that foreshadow outbreak emergence more reliably than any single day’s snapshot.

4.7 Final Modeling Approach and Outcome

4.7.1 Framing the Problem

At this stage of our modeling, the data had been reshaped into a fairly compact structure: for each country–day, we had a set of behavioral signals (stigma proportions, sentiment measures, topic prevalence) and a binary label indicating whether that day fell inside the 30-day pre-outbreak window (early_warning = 1) or not (early_warning = 0).

The task we ended up with was summarized as follows:

Given the emotional and topical state of online discourse on a given day, *what is the probability that this day belongs to the "pre-outbreak" period?*

This is essentially a binary classification problem, but with a heavy time dependence and a strong class imbalance since there are many more non–early-warning days than early-warning days. Rather than jumping straight into complex sequence models, we treated this as an opportunity to see how much signal could be extracted with a relatively simple, interpretable baseline: class-weighted logistic regression.

4.7.2 Feature Set

The input features were designed to capture both what people were talking about and how they were talking about it, as well as how those signals were evolving over time. Our final feature set was roughly as follows:

1. **Stigma proportions (current day)**
   a. stigma_fear : share of posts labeled as fear
   b. stigma_urgency : share labeled as urgency
   c. stigma_humor : share labeled as humor
   d. stigma_neutral : (used mainly for completeness)
2. **Sentiment polarity features**
   a. polarity_mean : mean sentiment polarity across all posts
   b. strong_neg_share : fraction of posts with strongly negative polarity (e.g., compound ≤ -0.6)
3. **Time-series derivatives (lags and momentum)**
   For each of the main signals (fear, urgency, humor, polarity), we included:
   a. 1-day, 3-day, and 7-day lags
      i. e.g., stigma_fear_lag3 = fear proportion 3 days ago
   b. 7-day rolling averages

        i. e.g., stigma_urgency_roll7 = average urgency over the last week

These were computed separately within each country so that the model learned changes relative to that country's own baseline, rather than global averages.

4. **Topic-level features**
   In some experiments, we also added:
   a. daily prevalence of a few "core" topics (e.g., human cases, animal outbreaks, policy debates, conspiracy narratives)
   b. and, in a more exploratory fashion, topic-specific sentiment (e.g., mean polarity within human-case topics)

These were particularly useful for interpreting results, even if they didn't always add significant performance improvements.

Overall, our objective was to give the model a short recent history of fear, urgency, humor, and negativity rather than a single snapshot, so it could recognize patterns like "fear has been slowly climbing for a week" rather than just "fear is high today."

4.7.3 Model choice and training strategy

Our model of choice was a logistic regression classifier, for three reasons:

1. It is transparent and explainable. Coefficients and SHAP values can be interpreted in a straightforward way, which is important when claiming detection of an early-warning signal.
2. It is fast to train, even with thousands of country-day observations and dozens of derived features.
3. It provides a sensible probability output that can be calibrated =for different operational settings.

Because early-warning days are relatively rare, we enabled class weighting (class_weight="balanced"), which up-weights positive examples in the loss function so the model doesn't simply learn to predict 0 each time. Features were standardized (zero mean, unit variance) before fitting.

For evaluation, we used:

- A simple train–test split that preserved class balance (via stratification), and
- Metrics that are appropriate for imbalanced problems:
  - **AUROC** (Area Under the ROC Curve)

- **PR-AUC** (Area Under the Precision–Recall Curve)

In a more extended deployment, stricter schemes (time-based splits, country-level cross-validation) would be preferrable; however, under the limitations of this study, our goal was to investigate whether there is *any* robust predictive signal at all.

Chapter 5: Findings, Conclusions, and Implications

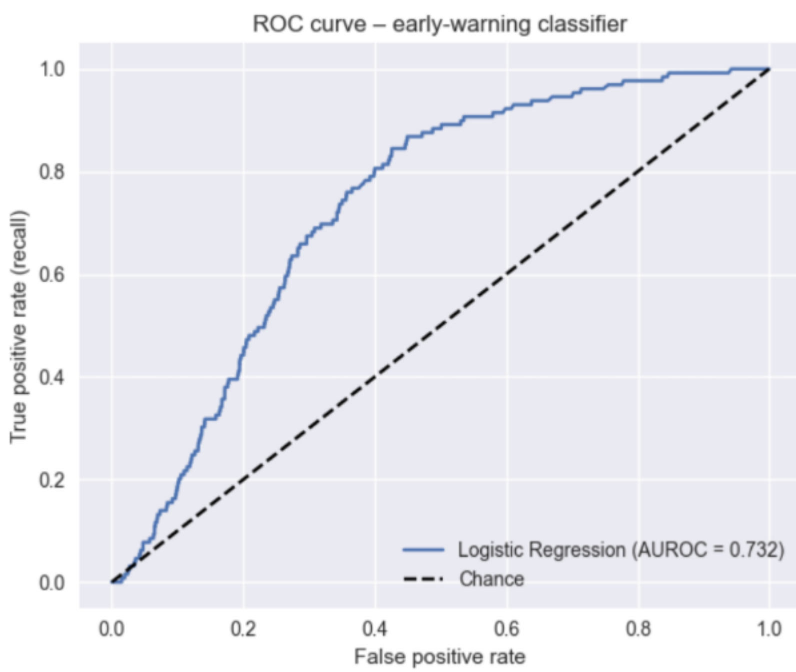
5.1 Findings

5.1.1 Model Performance and Early-Warning Signal Detection

The logistic regression model demonstrated that digital discourse around H5N1 contains a detectable early-warning signal, though embedded within considerable noise.

On the held-out dataset:

- **AUROC ≈ 0.73**, indicating the model ranks early-warning vs. non-warning days substantially better than chance.
- **PR-AUC ≈ 0.03**, which appears low but is meaningfully above the underlying positive base rate (~1–2%).



At the **default 0.5 threshold**, the model behaved as follows:

- For the negative class (0), precision was very high (~0.99) but recall was around 0.58.
- For the positive class (1), recall was quite high (~0.82), but precision was very low (~0.03), yielding a small F1.

In other words, when we use a standard threshold, the model is willing to flag most early-warning days (good recall) but does so at the cost of many false alarms, which pulls precision down.

We then searched for a threshold that better balanced things for the early-warning class. The "optimal" threshold in this simple sweep was around **0.52**. At that point:

- Positive-class recall dropped to about **0.64**,
- Positive-class precision rose slightly to about **0.04**,
- Overall accuracy improved to roughly **0.72**, and
- AUROC and PR-AUC of course remained the same (threshold-independent).

This is still a very imbalanced regime, so no threshold will magically give both high precision and high recall. Instead, the model behaves like a probabilistic filter: it can push outbreak-adjacent periods upward in the ranking, but any operational system would need additional layers (e.g., minimum support across multiple days, or cross-validation with other data streams) before acting on the signal.

The sustained upward trajectory in urgency corresponds closely with the theoretical background on digital epidemiology. Prior studies show that online behavioral signals are more informative when they accumulate steadily over time rather than emerging as short lived spikes. The cumulative rise in urgency identified in this study reflects heightened information seeking pressure and indicates that such behavioral shifts may serve as early digital indicators preceding formal outbreak recognition.

5.1.2 Model Explainability

To move beyond aggregate performance metrics and further understand how the early-warning model is behaving, we turned to SHAP (SHapley Additive exPlanations). SHAP is a model-agnostic framework that attributes a prediction to its input features in a principled way. For each country–day, it decomposes the model's output into a baseline term (what the model would predict with no information) plus a set of feature contributions that either push the prediction up toward "early warning" or pull it back down toward "no warning." Because these contributions are grounded in Shapley values from cooperative game theory, they add up cleanly to the final prediction and are directly comparable across features.

In practice, this means we can do two things that standard coefficients alone do not give us. First, at a global level, we can see which features consistently have the largest impact on predictions, and whether high values of those features tend to increase or decrease risk. The SHAP summary plot provides exactly this view: each point represents a single observation, coloured by the feature value and positioned according to how much it changed the model's output. Patterns in these clouds reveal not only importance, but also directionality and non-linear behavior. Second, at a local level, we can inspect individual days and countries to see how specific combinations of features drove a particular early-warning probability. It is essentially a step-by-step explanation of why the model was "worried" on that day.

Using SHAP in this way is especially useful for our setting, where the inputs (fear, urgency, sentiment, and topic dynamics) are abstract behavioral signals rather than concrete biological measurements. The method allows us to check that the model's internal logic matches our substantive understanding of outbreak dynamics: that sustained urgency and multi-day fear should matter more than one-off spikes, and that topic context should modulate the strength of those signals. Therefore, the SHAP summary plots serve as both a diagnostic tool and an interpretive bridge, translating a fitted logistic regression into an intelligible narrative about how digital discourse patterns relate to early-warning risk.

5.1.3 SHAP Summary Plot

The summary plot (Figure 7) presents a compact visual representation of how the logistic regression model is utilizing its main features. Each row corresponds to a single feature, and each point represents a single country–day in the test set. The horizontal position indicates how much that feature influenced the prediction and in which direction. The color gradient represents the value of the feature on that day, where pink/red signifies a high value and blue signifies a low value.

Starting at the top, urg_roll7 clearly dominates the plot. Its cloud stretches farthest to the right, meaning that for some days the seven-day rolling urgency alone can push the model more than a full logit unit toward an early-warning prediction. The color gradient is very clean: when urg_roll7 is high (the red points), the SHAP values are almost always positive; when it is low (blue), the SHAP values cluster at zero or slightly negative. In other words, a week of elevated urgency language reliably increases the early-warning probability, whereas a week of calm, low-urgency discourse dampens the signal.

Below that, fear_lag3 and fear_lag7 show a similar but slightly weaker pattern. High values of fear three or seven days ago tend to sit on the right side of the plot, nudging predictions upward, while low values sit on or just left of zero. The horizontal spread is smaller than for urg_roll7, which tells us that multi-day fear is important but not as decisive as multi-day urgency. The model is effectively telling us: "if people have been fearful in the past few days, that matters, but it matters most when it coincides with sustained urgency."

fear_roll7 looks slightly different. The points are more densely packed around zero, and the color gradient is more mixed. This suggests that the seven-day rolling fear signal is being used more as a fine-tuning feature. High rolling fear sometimes pushes the prediction up, but the effect size is modest; low rolling fear sometimes pulls it down, but not dramatically. This is consistent with a model that already has several fear lags: the roll-7 statistic adds some smoothing and context, but the lags themselves carry much of the information.

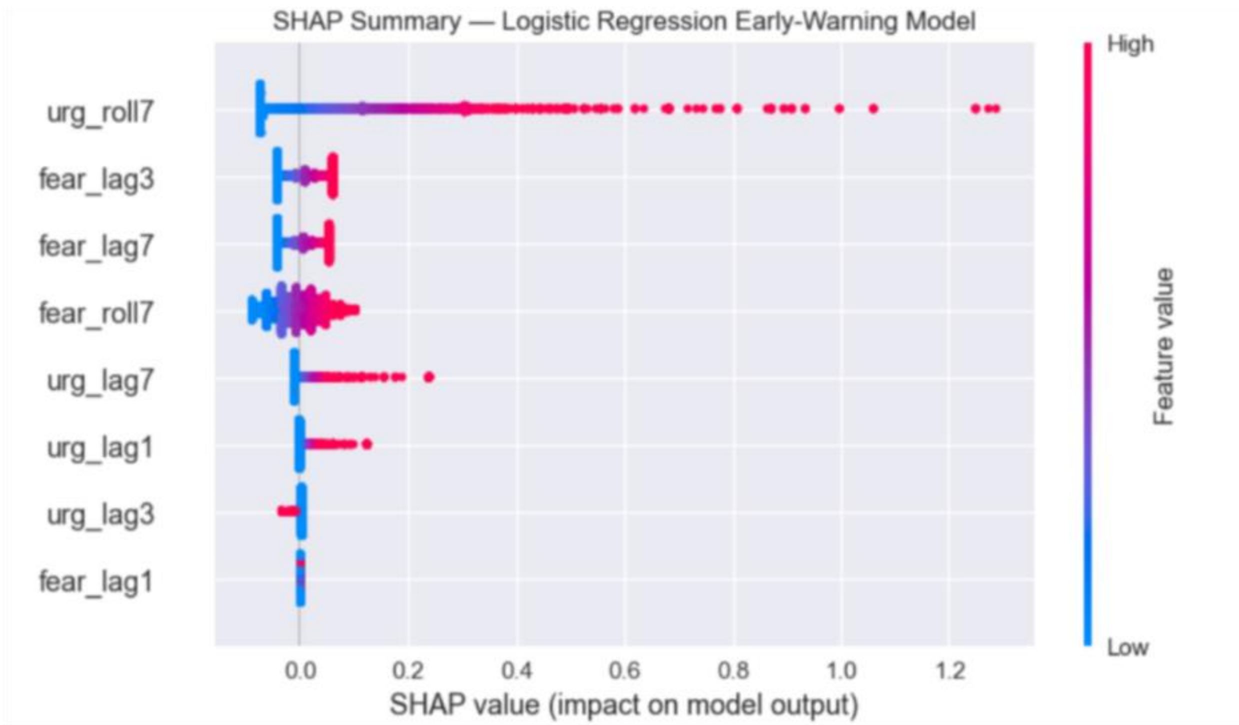


Figure 7: SHAP Summary Plot

Lower down, the urgency lags (urg_lag7, urg_lag1, urg_lag3) have noticeably smaller spreads, and their points cluster more tightly around the vertical axis. They still show the expected pattern—higher urgency yesterday or three days ago tends to nudge the model toward early warning—but their influence is mostly incremental. They adjust the prediction around the behavior of urg_roll7 rather than overturning it. fear_lag1 at the bottom is the weakest of the set: the points are very close to zero, and both red and blue dots overlap heavily, indicating that a one-day spike or dip in fear by itself rarely changes the model's behavior.

Overall, the plot reveals a very coherent pattern. The model prioritizes patterns over isolated blips. A sustained run of urgent language over the last week, especially when accompanied by elevated fear in the last few days, is what consistently pushes days into the high-risk regime. When urgency and fear remain low or fluctuate only briefly, the SHAP values sit near zero and the model's output stays close to the baseline. This behavior aligns closely with the desired outcome of an early-warning system built on noisy social signals. The model does not chase individual minor fluctuations but rather responds strongly when concern and urgency build and persist over time.

5.1.4 Feature Importance

The feature-importance bar plot in Figure 8 provides a direct ranking of what the early-warning model leans on most often. Each bar represents the average absolute SHAP value for a feature. A longer bar corresponds to a feature that routinely has a larger impact, either positive or negative, on the predicted early-warning probability.

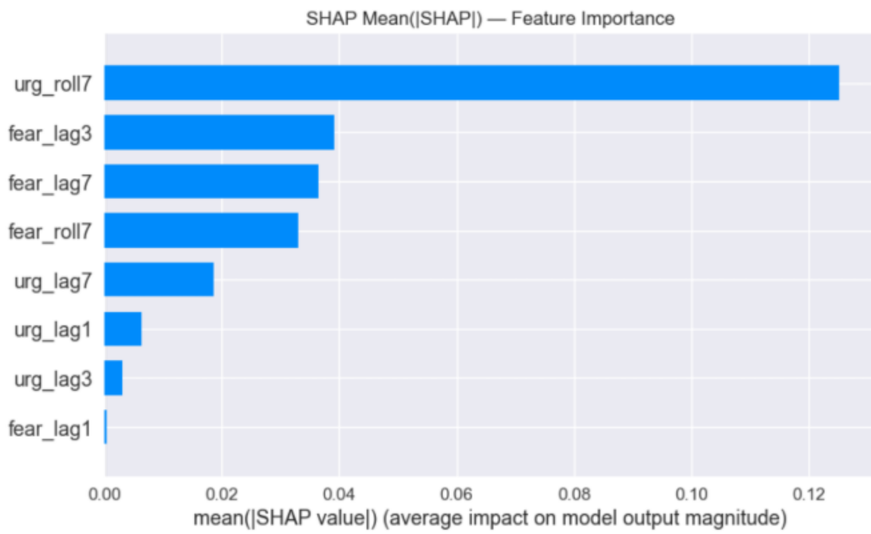


Figure 8: Feature Importance Bar Plot

The standout feature on the plot is the seven-day rolling urgency, urg_roll7. Its bar is substantially longer than all the others, which indicates that this single measure of sustained urgency contributes more to the model's decisions than any other variable across the test set. In practical terms, the model is first and foremost asking: "Has urgency been high over the last week?"

Immediately below the top feature, a cluster of fear-related features forms the second most significant group of predictors: fear_lag3, fear_lag7, and fear_roll7. Their bars are shorter than urg_roll7 but still clearly separated from the rest. Fear three days ago, fear a week ago, and average fear over the past week are all important factors, which reinforces the principle that a multi-day pattern of elevated concern is more informative than a one-day spike.

The urgency lags (urg_lag7, urg_lag1, and urg_lag3) reside in a lower tier. They are not irrelevant, but their average impact is noticeably smaller. They appear to fine-tune the risk estimate influenced by urg_roll7 rather than driving the prediction on their own. Finally, fear_lag1 exhibits a notably small bar, suggesting that isolated day-to-day fluctuations in fear contribute very little compared to the broader temporal patterns.

Overall, the bar plot reinforces the findings from the SHAP summary: the model's early-warning behavior is dominated by sustained urgency and multi-day fear, with shorter-term lags playing a modest supporting role.

5.2 Conclusions

In summary, our results suggest that there is in fact a non-trivial early-warning signal in digital stigma and sentiment, but it is noisy and asymmetric. The model is good at finding many of the early-warning days (high recall) and ranking them above background days (AUROC ≈ 0.73). However, precision remains low, which means any operational use would need to tolerate a fair number of false positives or layer this model with additional filters.

We consider this study to be a baseline proof of concept rather than a finalized forecasting tool:

- Periods leading up to outbreaks do show a distinct emotional and topical signature.
- Even a relatively plain logistic regression, fed the right temporal features, can detect that structure better than naive baselines like random guessing or a single fear threshold.
- The model's coefficients and topic-aware explanations give a concrete starting point for more sophisticated temporal models (e.g., sequence models, survival analysis, or hierarchical approaches that pool information across countries).

In short, digital stigma and sentiment signals are messy, but they are not random. When we aggregate them to the country–day level, add temporal context, and tie them to specific outbreak narratives, we can recover patterns that often precede formal outbreak recognition. The current logistic model is best viewed as a baseline early-warning layer that is interpretable, imperfect, and clearly improvable, but already capable of turning chaotic online chatter into structured, reproducible risk indicators.

5.3 Implications

This study demonstrates that emotional expressions on digital platforms can serve as early indicators of emerging infectious disease risks that often remain undetected by traditional surveillance systems. The findings show that sustained increases in fear and urgency are not merely reflections of heightened public interest, but rather representations of collective shifts in perceived risk that precede formal outbreak announcements. This suggests that digital epidemiology can function as a complementary layer of early warning that enhances the responsiveness of public health systems. Such emotional signals, when integrated with conventional epidemiological data, may offer a multi-dimensional decision framework for early risk assessment. These patterns indicate that collective emotional trajectories reflect underlying shifts in perceived vulnerability that precede institutional recognition of risk.

Furthermore, the study highlights that persistent urgency signals are closely associated with uncertainty and the amplification of misinformation. These signals represent moments in which institutional communication becomes particularly critical, as they reflect rising information demand and growing instability in the digital discourse environment. Public health agencies may therefore leverage emotional dynamics as indicators for when to intensify messaging or intervene in ongoing narratives. This insight contributes to a more precise understanding of how digital emotion patterns can support real-time risk communication strategies. By detecting these surges in information seeking, public health authorities can better anticipate when narrative interventions may mitigate harmful amplification dynamics.

The combined use of topic modeling and emotional analysis sheds light on how specific narratives become coupled with particular emotional reactions. Narratives involving poultry outbreaks, wild bird die offs, companion animal safety, and potential human spillover were consistently associated with more negative sentiment, indicating that certain topics exert stronger influence on public concern and risk amplification. The observed interaction between emotional signals and narrative structures aligns with the theoretical background on Information Disorder Theory and SARF. Periods of limited or uncertain official information heighten public information seeking behavior and create conditions under which speculative or misleading content can circulate more readily. At the same time, symbolically salient risk narratives such as human infection reports or mass wild bird die offs tend to pair with fear and stigma, thereby amplifying perceived risk. These findings demonstrate how information disorder and risk amplification processes materialize within digital environments and offer important implications for the development of targeted public health communication strategies.

Finally, the explainability analysis confirms that early warning signals are shaped more by sustained emotional dynamics than by short term fluctuations. This emphasizes the need to incorporate temporal persistence and emotional momentum into future early warning system designs. Overall, the study contributes to a novel framework that links digital emotional signals to public health risk assessment and offers empirical support for integrating digital discourse analysis into outbreak preparedness strategies. These insights underscore that digital emotion signals can play a substantive role in strengthening the anticipatory dimensions of outbreak preparedness.

5.4 Limitations

Despite offering meaningful insights, the study is subject to several limitations. First, the analysis relies primarily on English language data, limiting the extent to which emotional and cultural nuances in non-English speaking regions can be captured. This raises concerns regarding the generalizability of the findings, particularly in regions where digital engagement is high but linguistic patterns differ substantially. Additionally, platform specific demographic biases mean that digital signals may not fully represent the broader population.

Second, the rule-based approach to emotional classification is limited in its ability to detect nuanced or context-dependent expressions such as sarcasm, irony, or meme-driven communication. These forms of expression may distort the measurement of fear or urgency, leading to potential overestimation or underestimation of certain emotional signals. More sophisticated models would be required to address these linguistic complexities.

Third, the construction of the epidemiological ground truth relies on data sources that vary in reporting timeliness and verification standards. Differences between actual outbreak occurrence and the timing of official announcements create structural noise in the alignment between emotional signals and epidemiological events. This may have influenced the evaluation of the model's early warning performance.

Fourth, the study employs a linear logistic regression model which, although interpretable, lacks the capacity to fully capture nonlinear emotional dynamics, long term dependencies, and narrative transitions. As such, the predictive power of the model may be constrained by its limited representational capacity. Finally, the study is based on retrospective data analysis and has not yet been tested in a real time operational setting where streaming data, anomaly detection, and adaptive model updates are essential.

5.5 Future Work

Future research may expand the contributions of this study across several directions.

First, more advanced emotional and narrative classifiers should be developed. LLM-based models can capture complex emotional expressions, contextual shifts, and mixed affect more accurately than rule-based systems. Incorporating such models would likely improve the sensitivity and specificity of fear and urgency detection, thus strengthening early warning performance.

Second, future work should explore nonlinear time series models capable of representing long term dependencies and cumulative emotional effects. Transformer based sequence models, state space approaches, and survival analysis frameworks could provide a more sophisticated representation of temporal emotional structures. These methods may enhance the model's ability to detect subtle or gradual changes in preceding outbreaks.

Third, a multilingual and regionally diverse analysis would enhance the generalizability of this study. While this research encountered access limitations that precluded the incorporation of Korean social media data, which had been included in the initial analysis plan, South Korea represents a particularly valuable case study due to relatively high rates of social media usage rates and rapid digital response to public health events. Including Korean data in subsequent research will advance the generalizability of digital epidemiology across linguistic and cultural contexts.

Fourth, multimodal early warning frameworks should be developed to incorporate images, videos, memes, and hashtag networks in addition to text. Public reactions to infectious disease events are shaped through diverse media formats, and multimodal analysis would provide a more comprehensive understanding of digital risk perception.

Finally, real world integration with public health agencies is essential. Operationalizing the model requires research on threshold selection, false positive cost management, and communication workflow integration. Conducting pilot tests with real time streaming data will be a crucial step toward translating the model into a functional component of outbreak preparedness and response.

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